

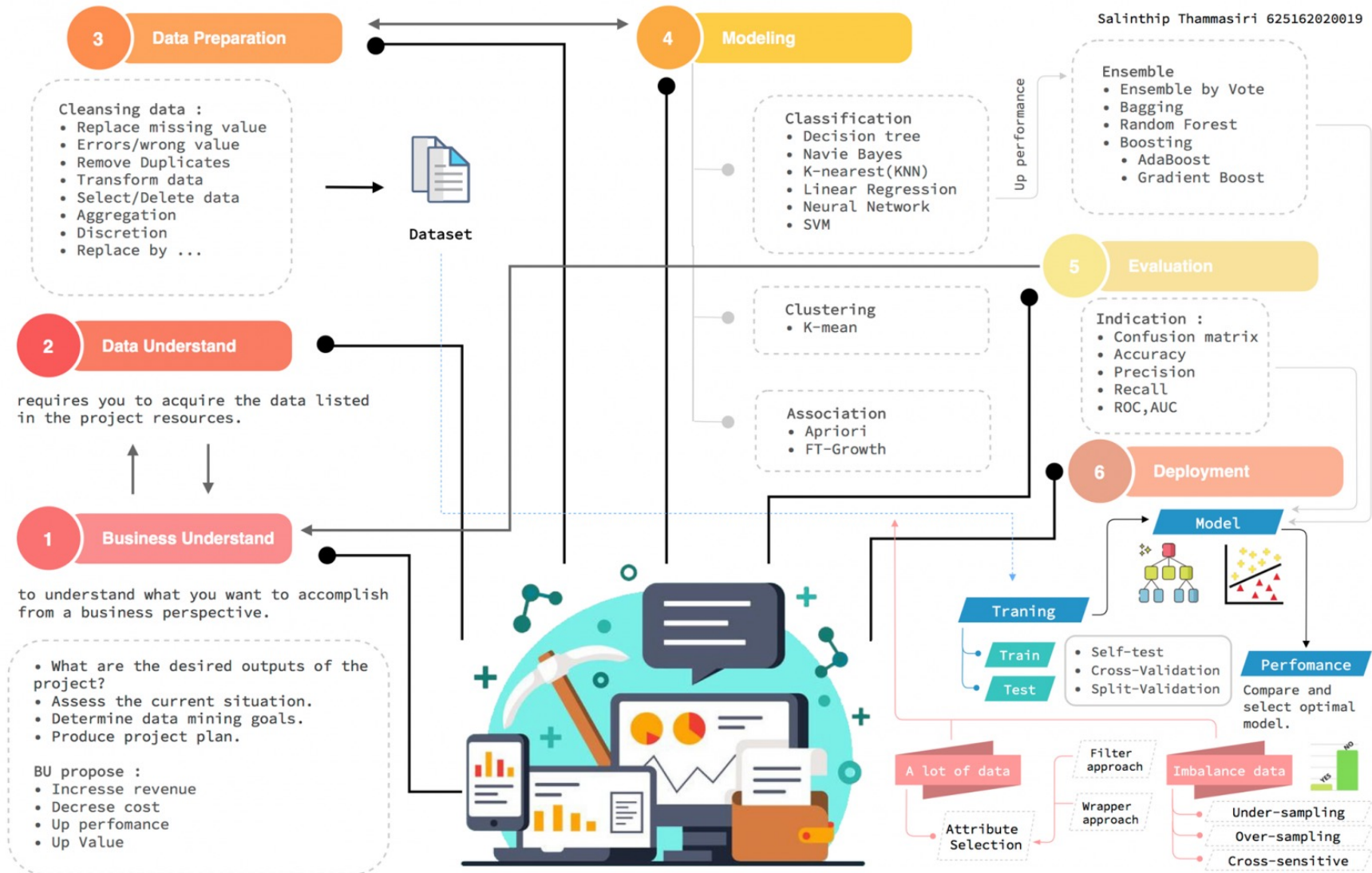
ดำรงฤทธิ์ เศรษฐศิริโชค

Deployment

- Deployment Concept
- MLOps
- WEKA Model Deployment
- Production Model Deployment
- Rapidminer Model Deployment
- Data-Centric AI

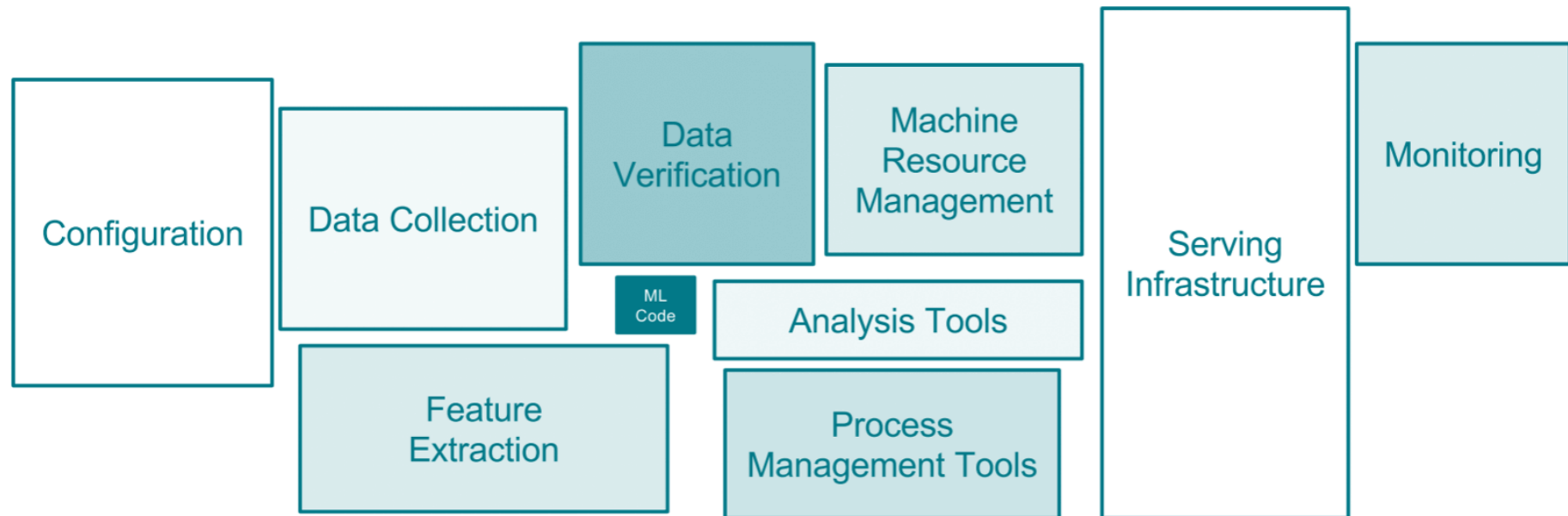


Cross-Industry Standard Process for Data Mining



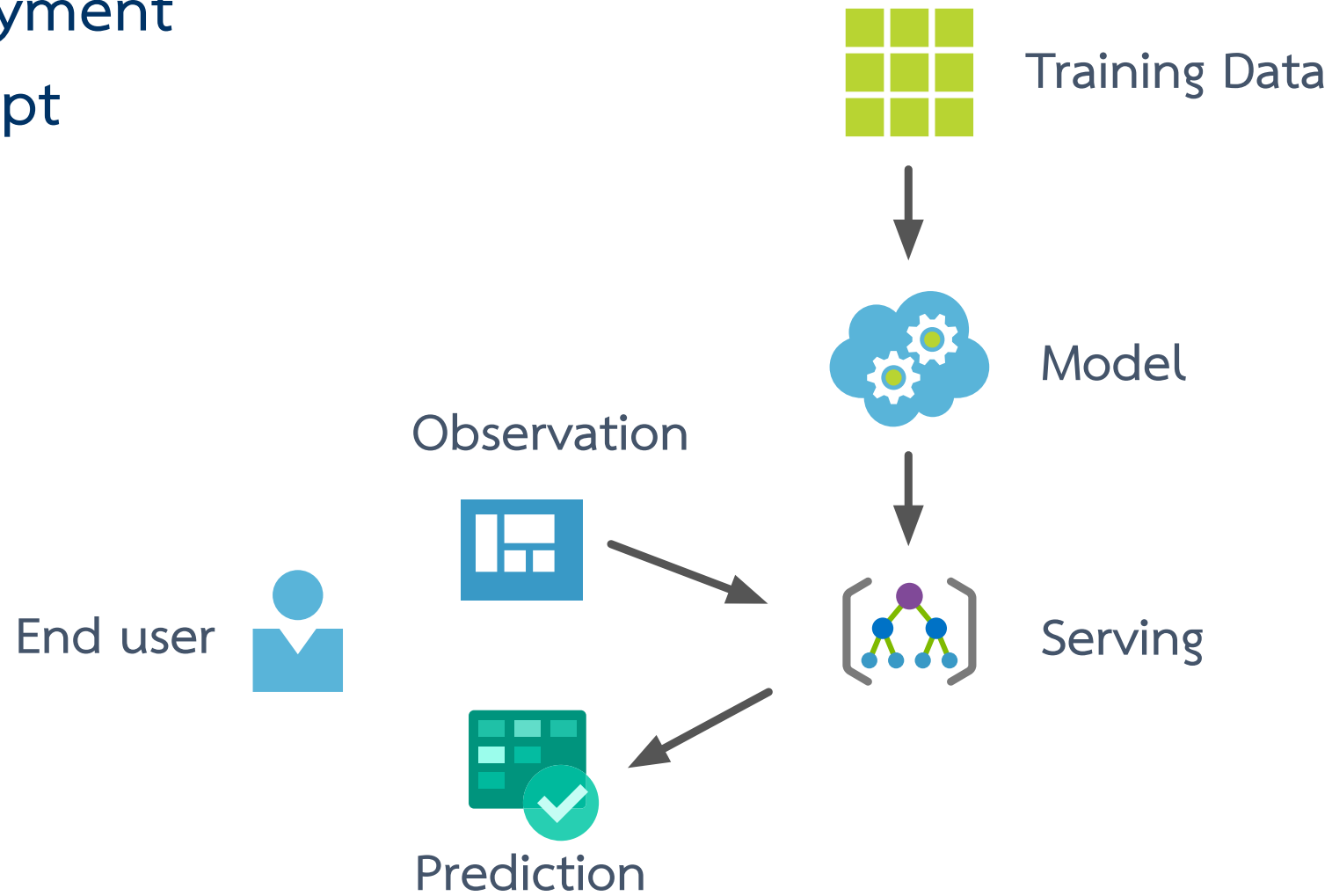
Hidden Technical Debt in Machine Learning Systems

Requirements for getting ML deployed

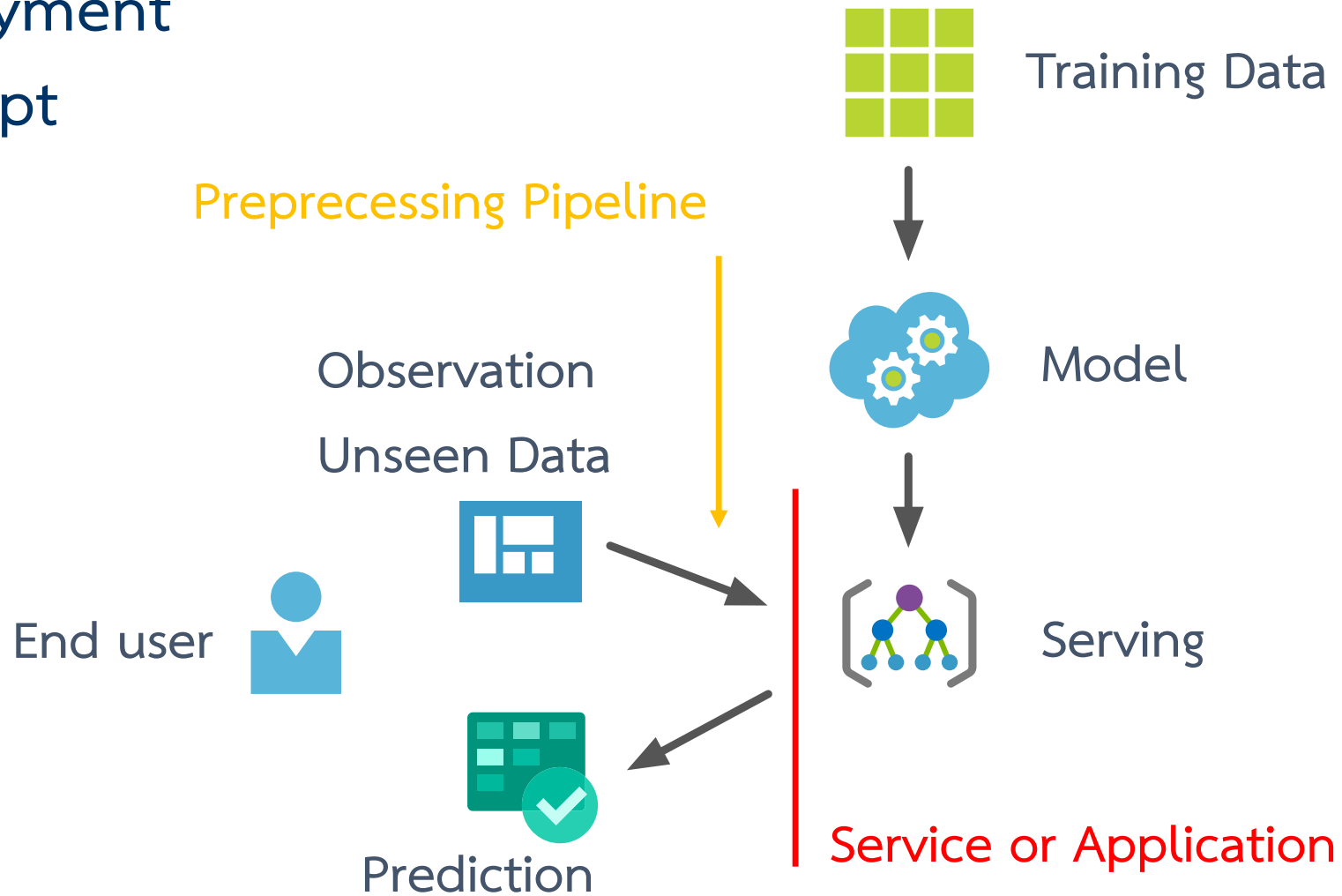


Ref: <https://proceedings.neurips.cc/paper/2015/file/86df7dcfd896fcdf2674f757a2463eba-Paper.pdf>

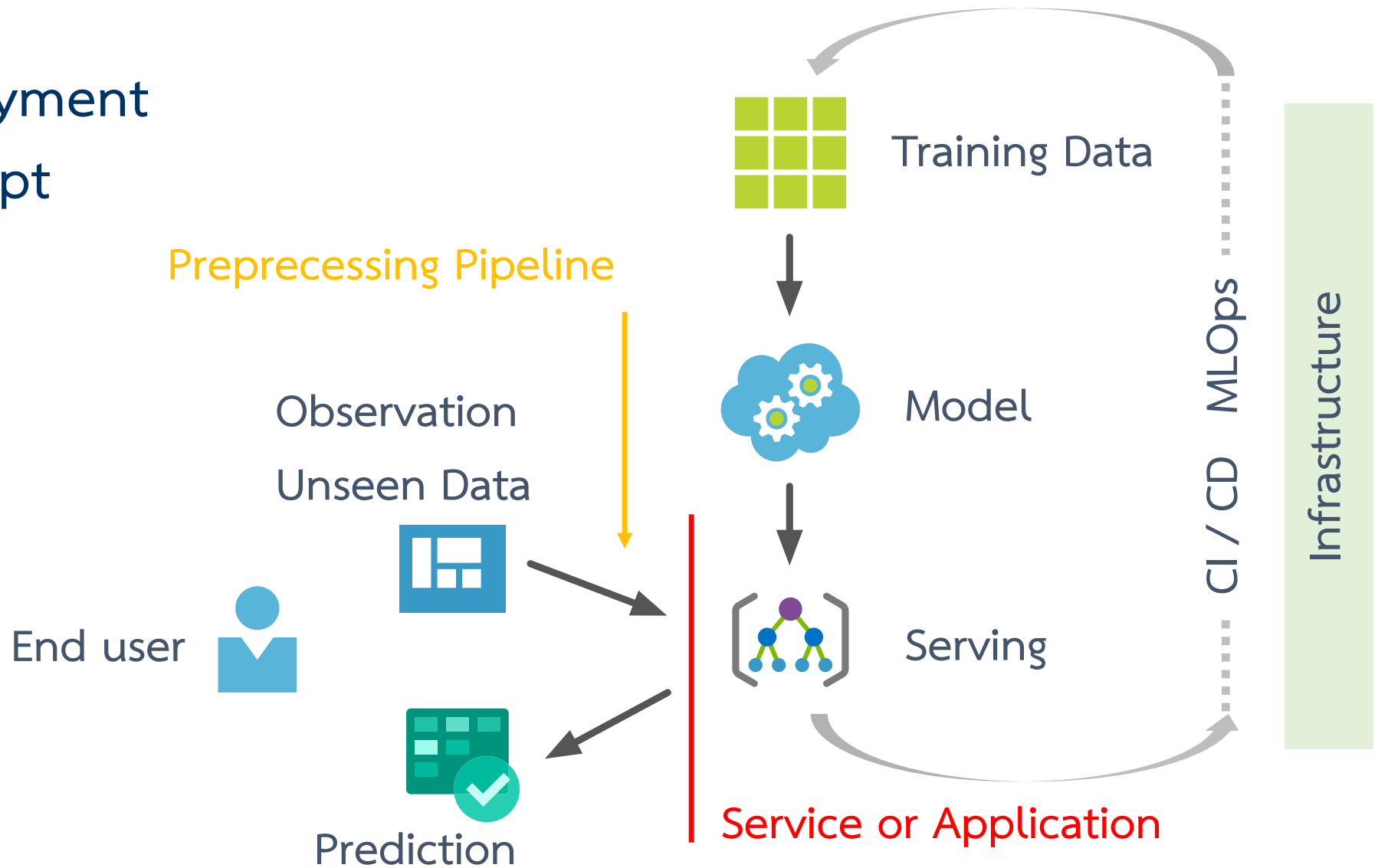
Deployment Concept



Deployment Concept



Deployment Concept



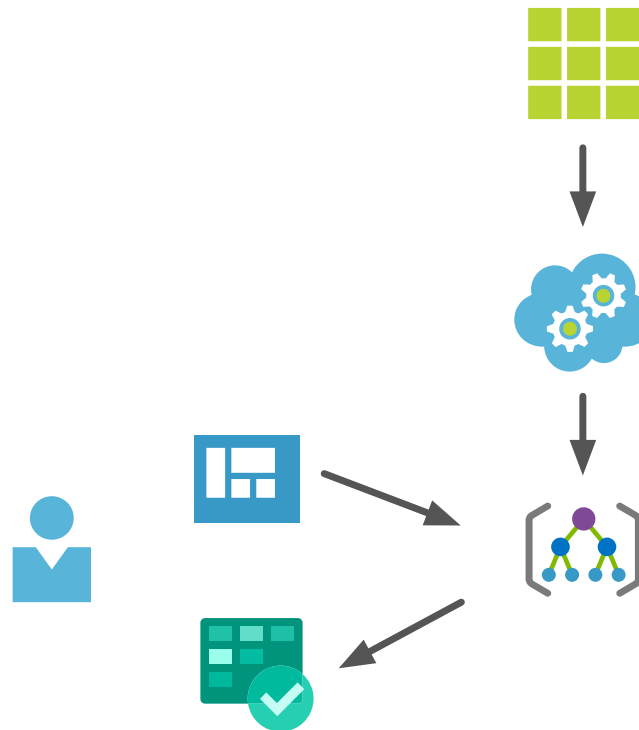
High-level of ML Systems

1 Data

- Data Engineering Pipelines
- Data Quality is a key

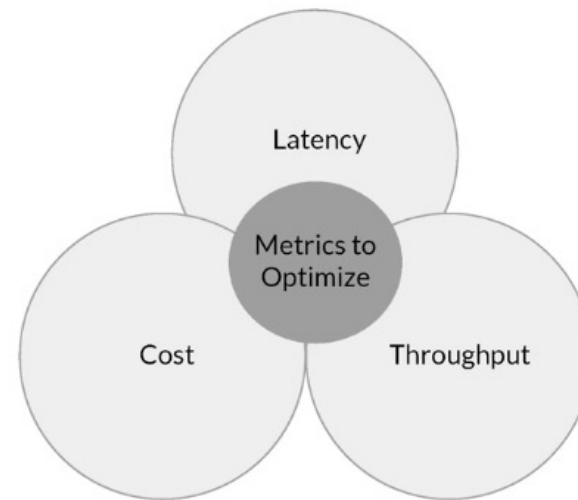
2. Model

- ML Pipelines
- Train, Evaluate, Test



Service or Application

- PLC
- Machine
- Batch Prediction
- Excel
- API
- Application Server
- Cloud
- Mobile
- Edge computing



Preprocessing Pipeline

- **Data Cleansing**
- **Feature Tuning:**
Dealing with missing values or outliers, Normalization
- **Feature Construction:**
Some prior calculation, Combine inputs, Feature crossing
- **Representation Transformation:**
Formatting, Encoding, Vectorization

CI/CD



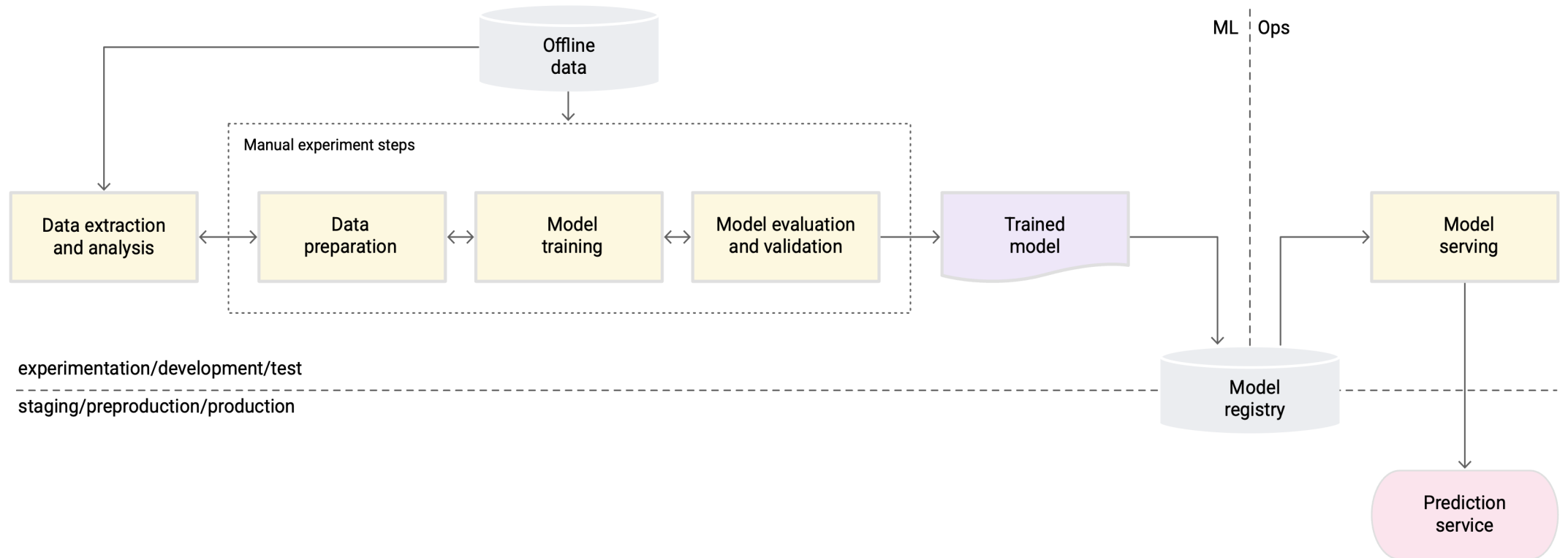
- Continuous Integration and Continuous Delivery or Continuous Deployment
- Version Control
- DevOps / MLOps (Machine Learning Model Operationalization Management)
- Logging and Monitoring Model
- Automate



Topic

- Deployment Concept
- **MLOps**
- WEKA Model Deployment
- Python Model Deployment
- Rapidminer Model Deployment
- Data-Centric AI

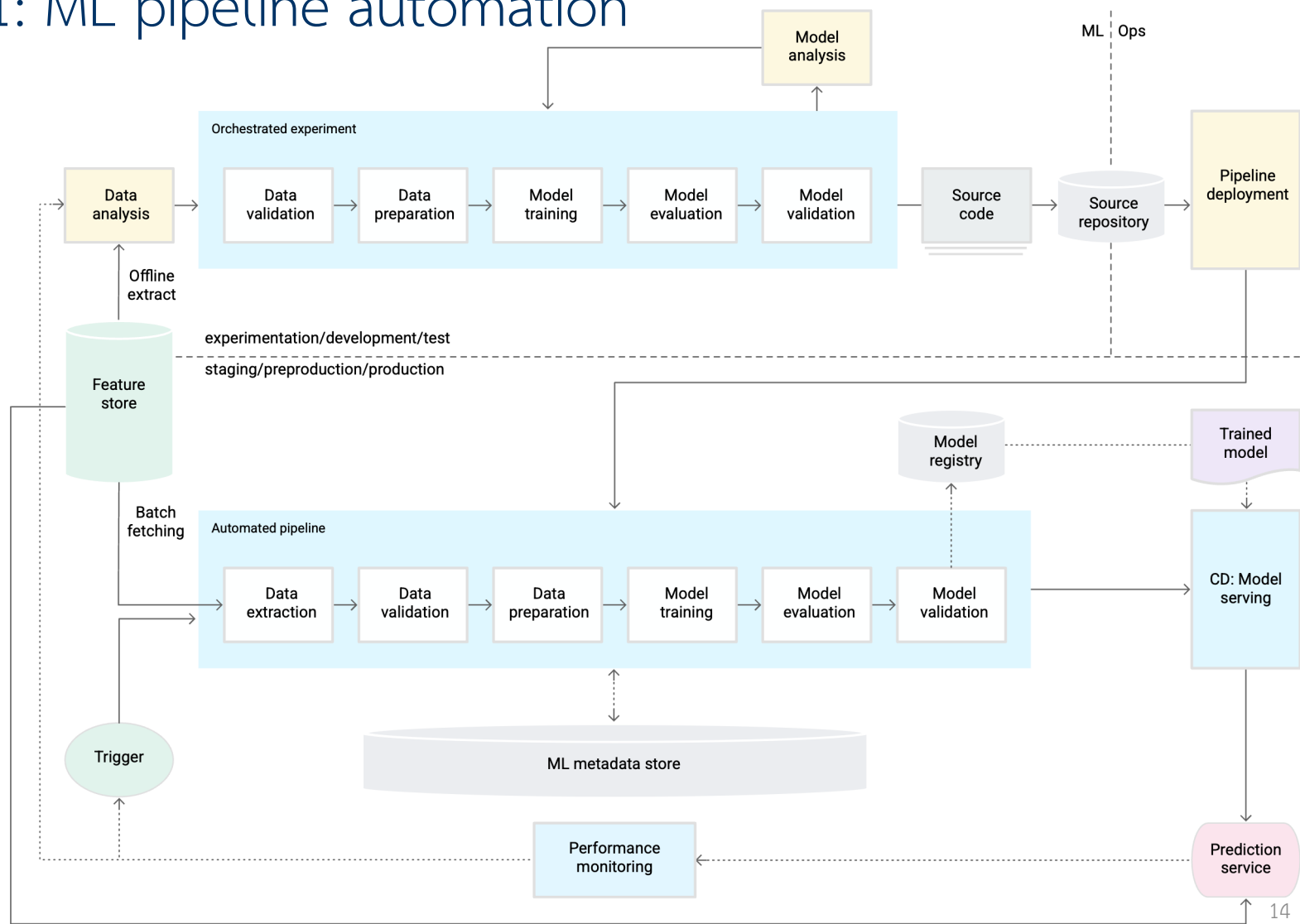
MLOps Levels 0



Manual ML steps to serve the model as a prediction service.

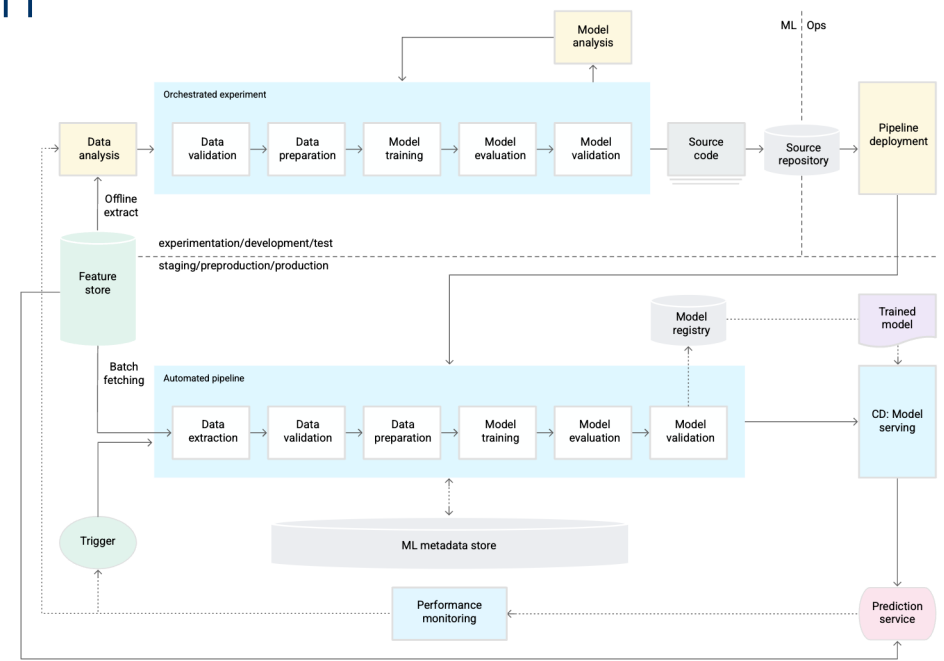
[Source: cloud.google.com](https://cloud.google.com)

MLOps level 1: ML pipeline automation



MLOps level 1: ML pipeline automation

- Data validation:
 - Data schema skews
 - Data values skews
- Model validation:
 - Comparing the evaluation metric
 - Consistency with the prediction service API
- Feature store:
 - Experimentation
 - Continuous training or online prediction



MLOps level 1: ML pipeline automation

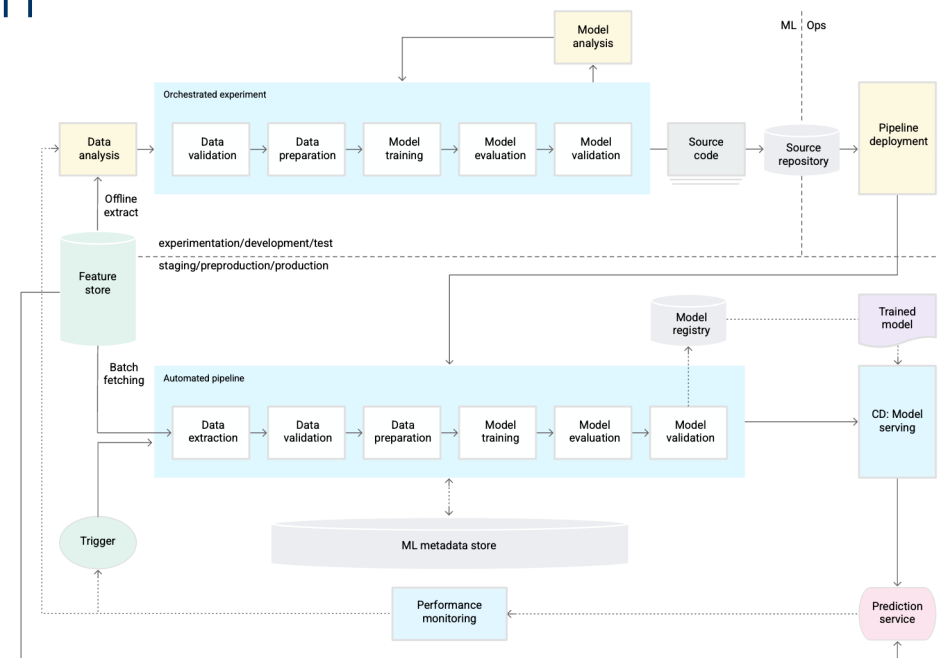
- Metadata management:

- Parameter arguments
- Model version control

- ML pipeline triggers:

- On demand
- Schedule

- *On model performance degradation*
- *Significant changes in the data distributions*



➡ Model monitoring

Model monitoring example

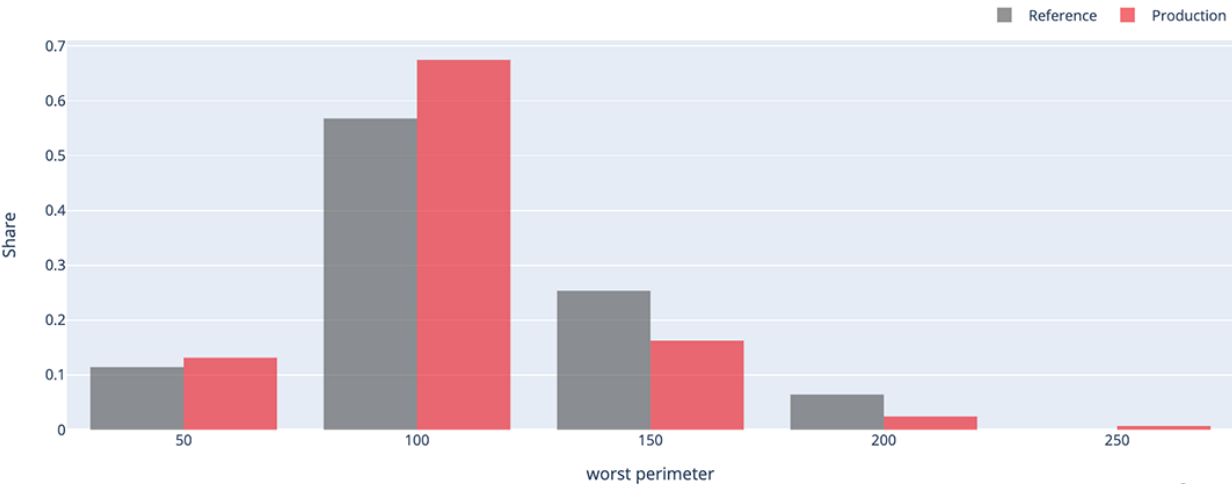
Data Drift Table

Q Search X

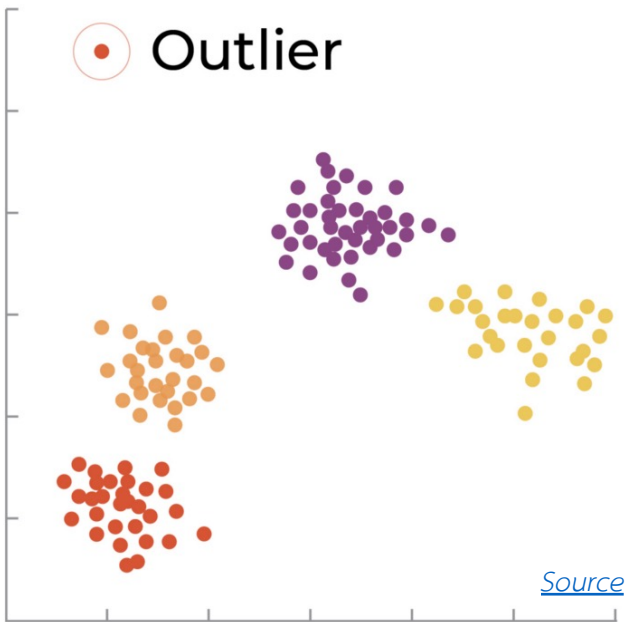
Feature	Type	Reference Distribution	Production Distribution	Data drift	P-Value for Similarity Test ↑
> worst concave points	num			Detected	0.000002
▼ worst perimeter	num			Detected	0.000005

DATA DRIFT

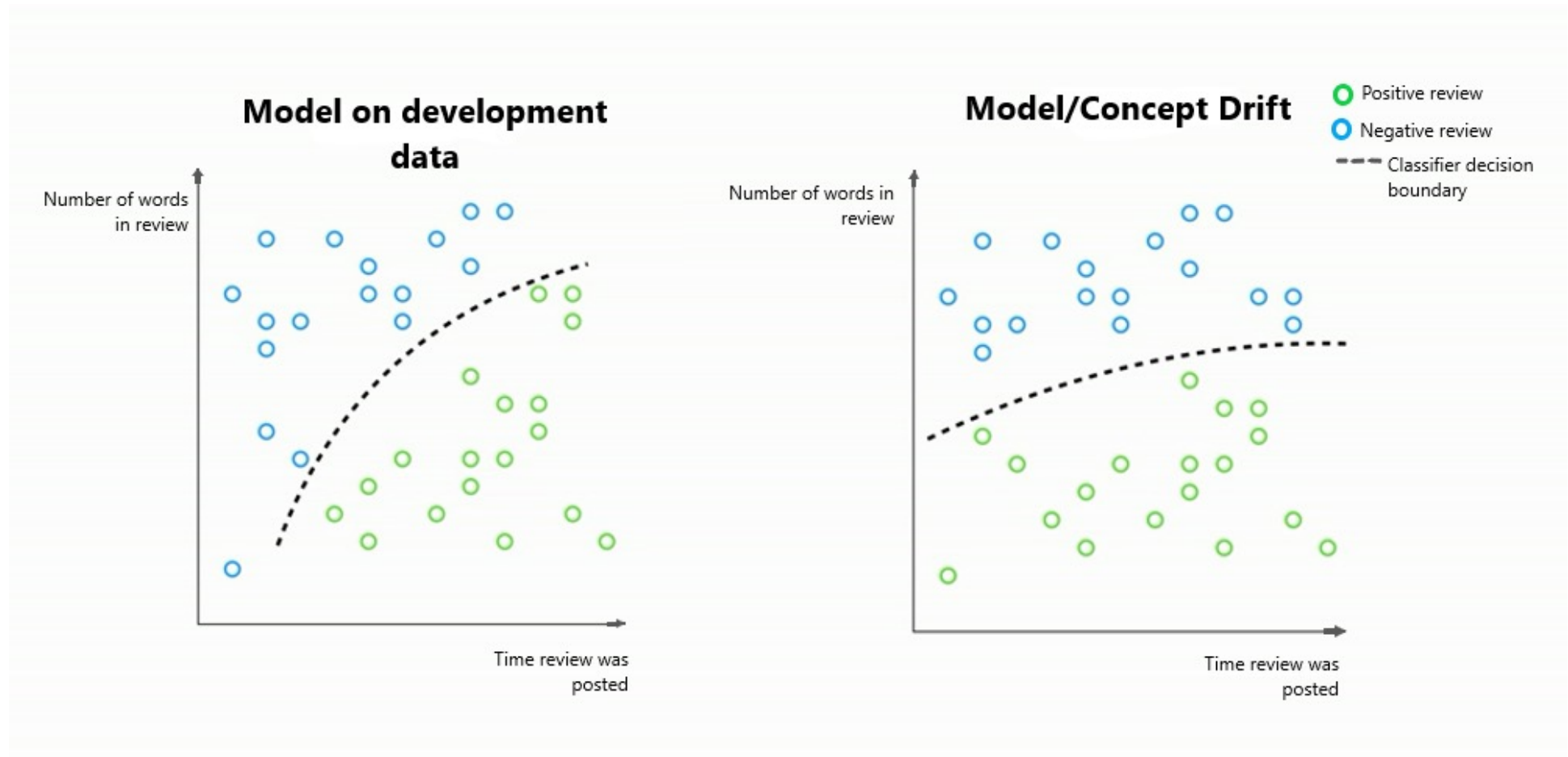
DATA DISTRIBUTION



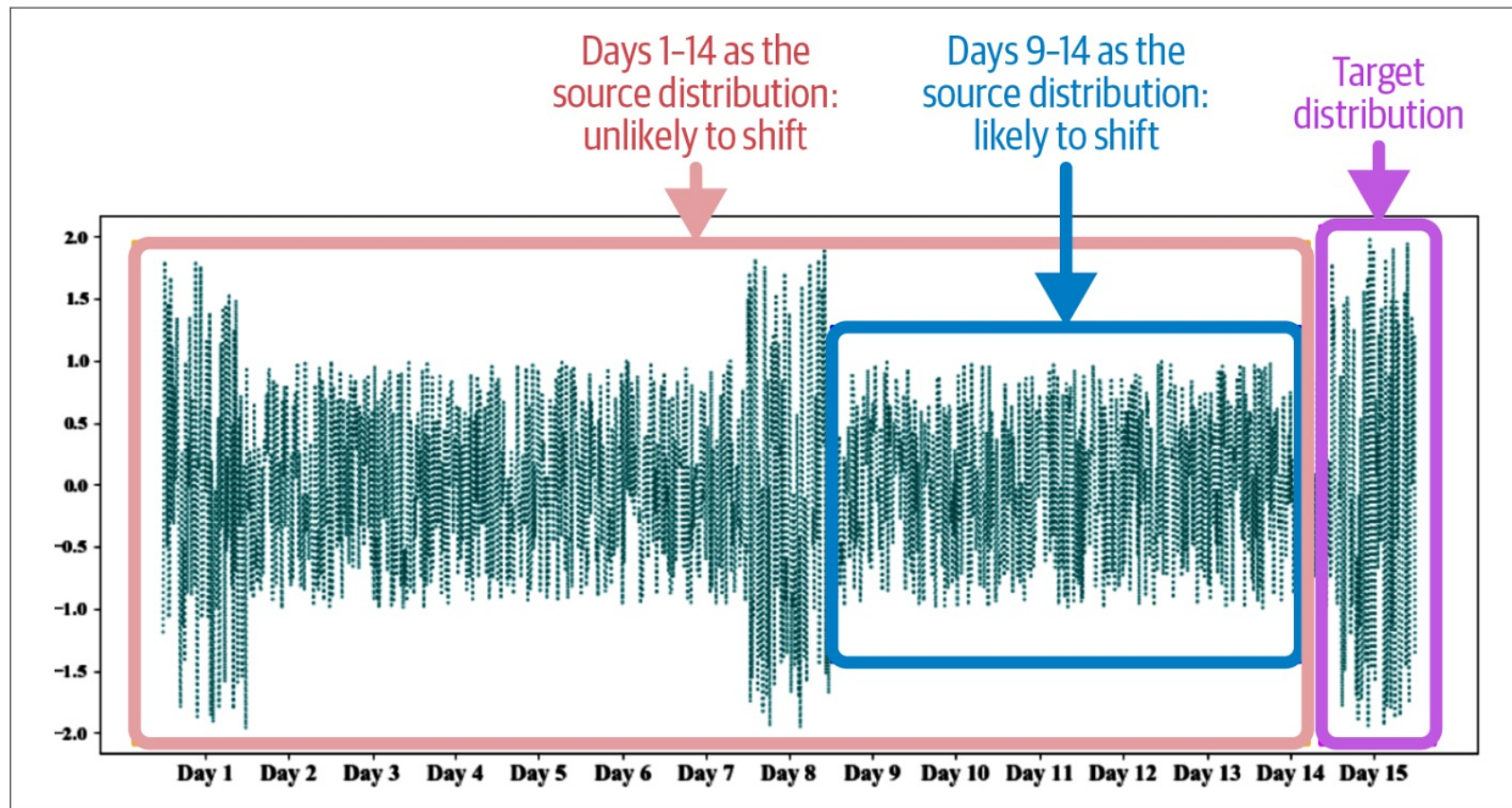
[Source](#)



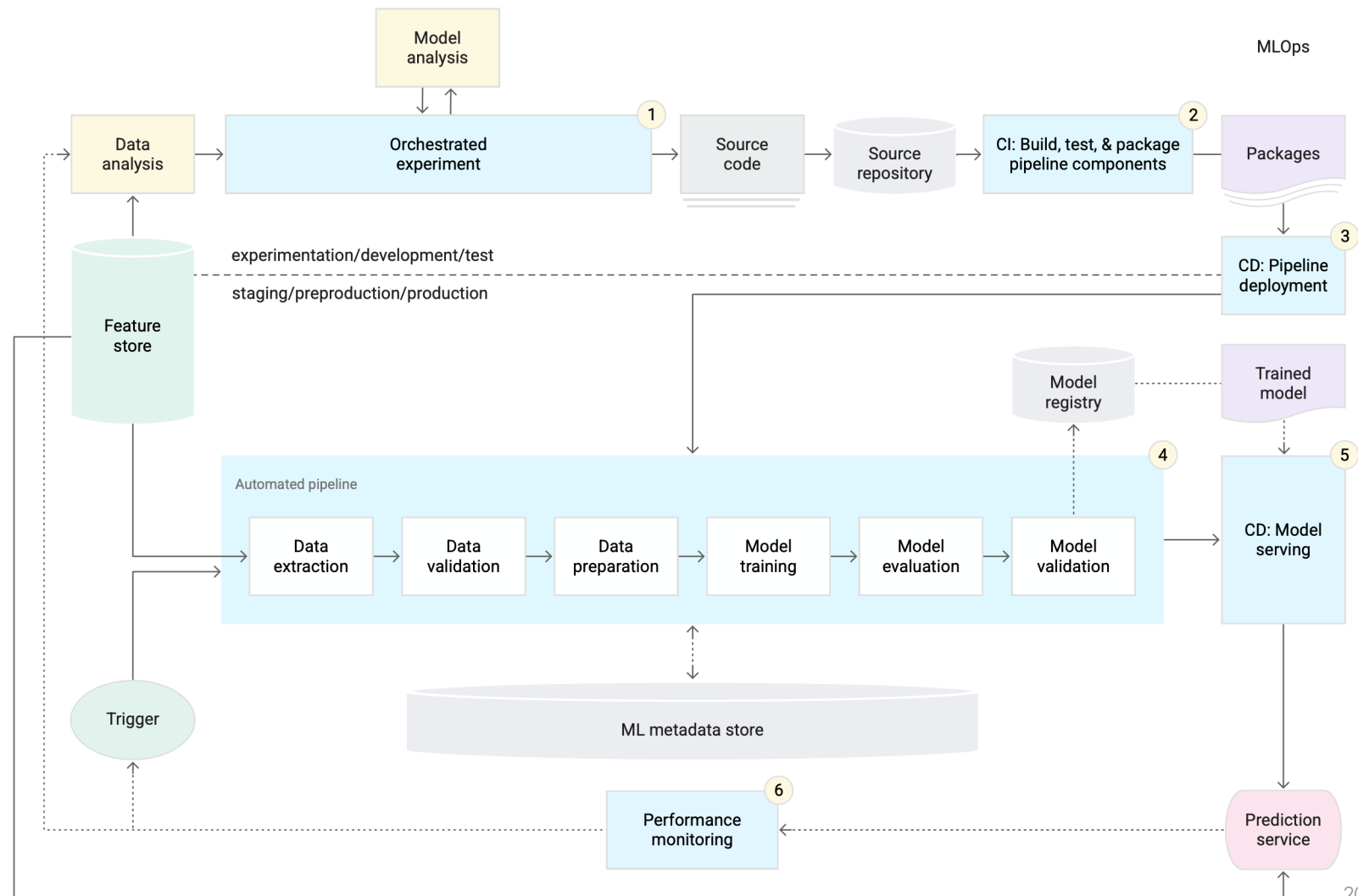
Model monitoring example



Detecting Shifts

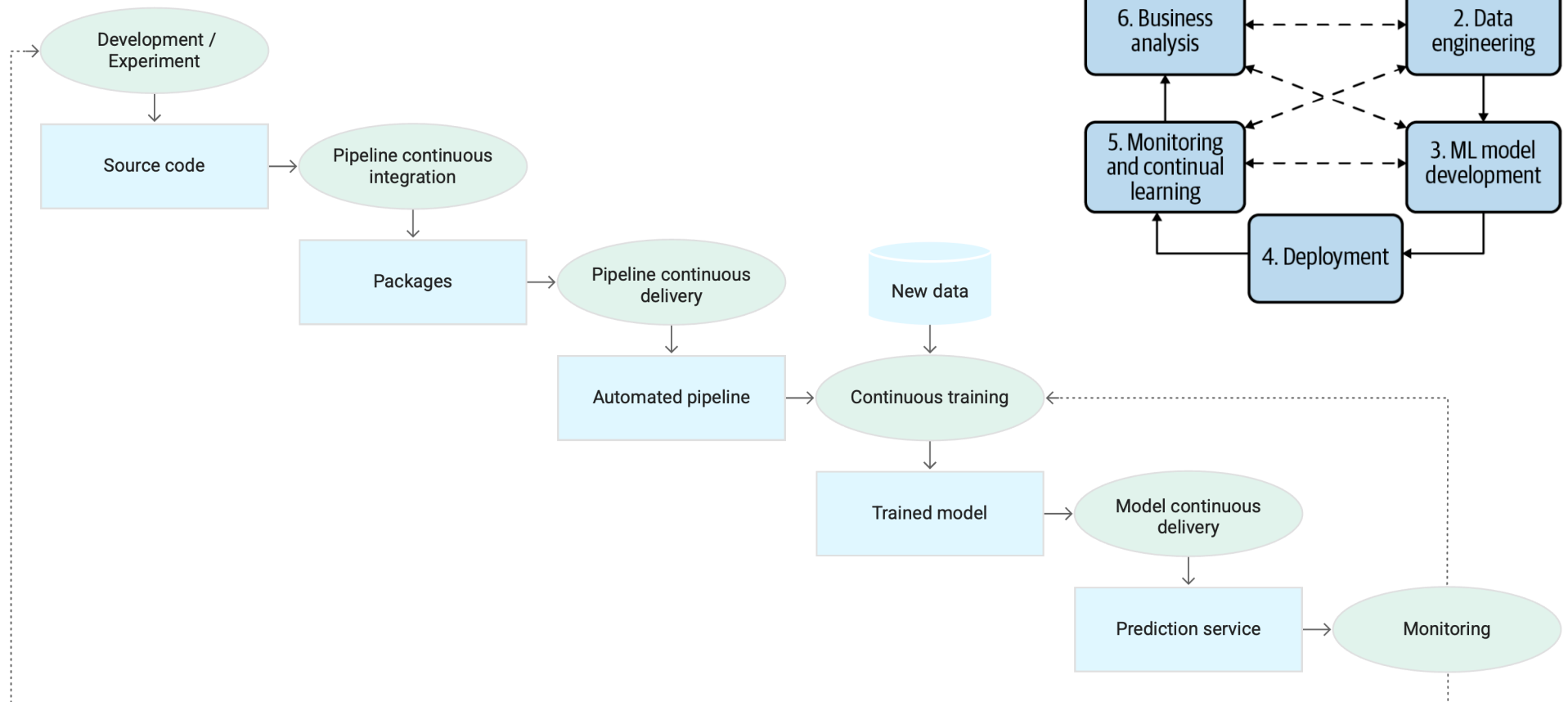


MLOps level 2: CI/CD pipeline automation



Source: [cloud.google.com](https://cloud.google.com/mlops)

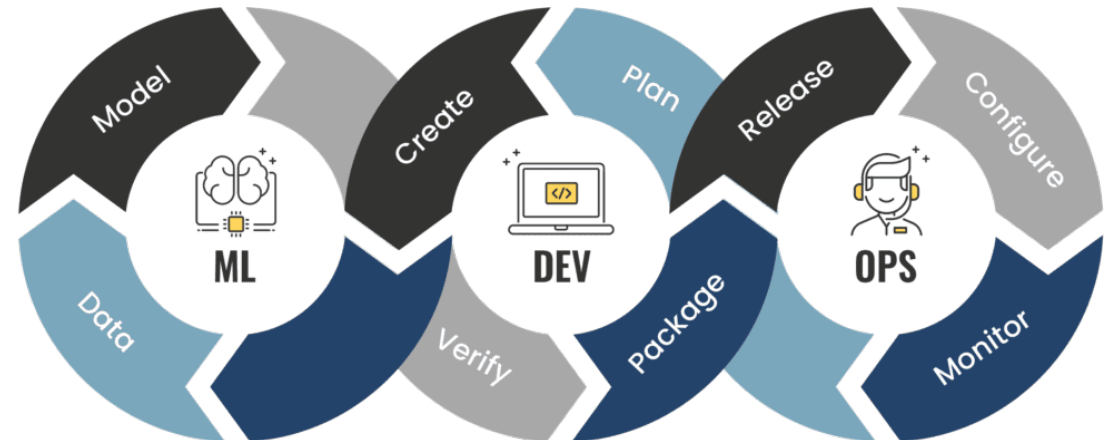
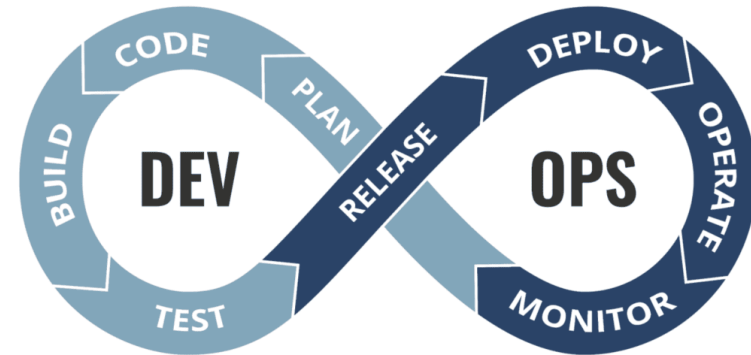
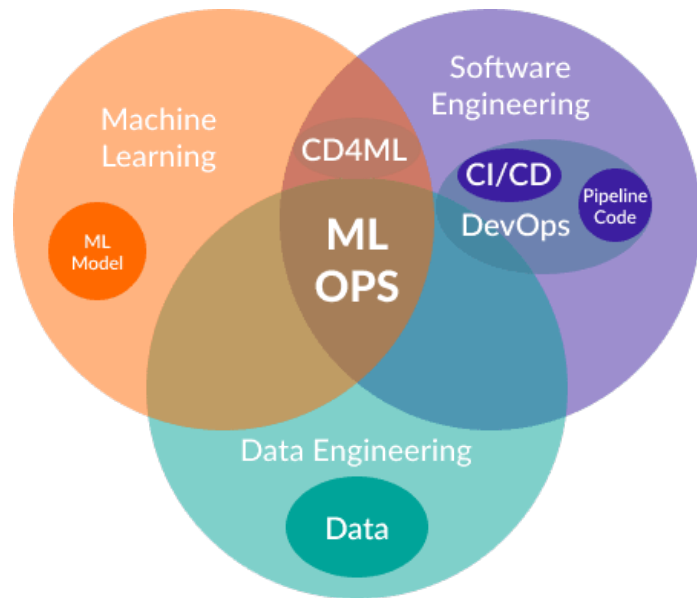
Stages of the CI/CD automated ML pipeline



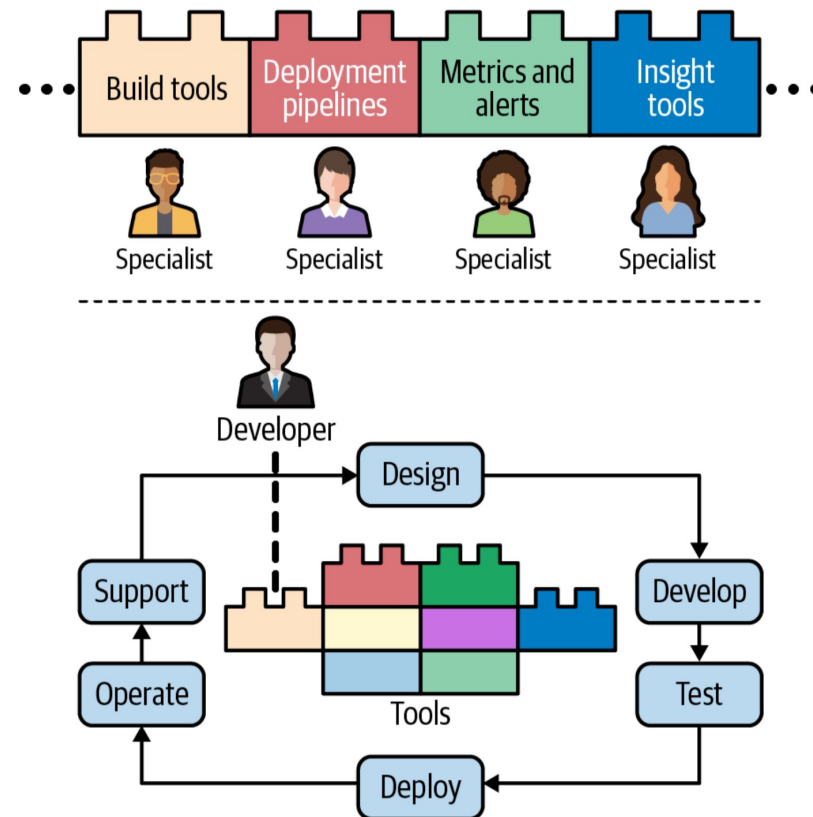
DevOps versus MLOps

- **Team skills:** The team usually includes **data scientists** or **ML researchers**, who focus on **exploratory data analysis**, **model development**, and **experimentation**.
- **Development:** Try different features, algorithms, modeling techniques, and **parameter configurations** to find the best for the problem as quickly as possible. Tracking what worked and what didn't, and maintaining reproducibility while **maximizing code reusability**.
- **Testing:** Need data **validation**, trained model **quality** evaluation, and model **validation**.
- **Deployment:** **Isn't as simple as deploying** an offline-trained ML model as a **prediction service**. Require to deploy a **multi-step pipeline** to automatically **retrain and deploy** models.
- **Production:** ML models can **decay** in more ways than conventional software systems, and you need to **consider this degradation**. Track **summary statistics**

DevOps versus MLOps



Example: Full-cycle developers at Netflix



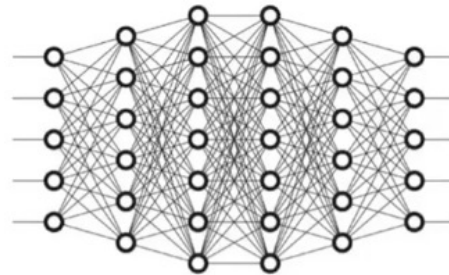
"Full Cycle Developers at Netflix—Operate What You Build," Netflix Technology Blog, May 17, 2018, <https://oreil.ly/iYgQs>.

Topic

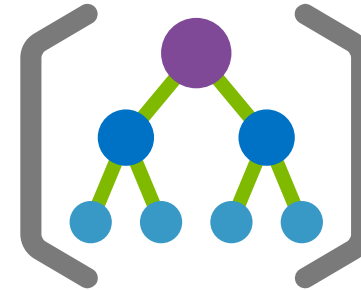
- Deployment Concept
- MLOps
- **WEKA Model Deployment**
- Python Model Deployment
- Rapidminer Model Deployment
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What is a Model ?

- Decision Tree
- Naïve Bayes
- Linear Regression
- Logistic Regression
- NN
- Deep Learning



$$p(c_i | x) = \frac{p(x | c_i)p(c_i)}{p(x)}$$



Attribute	Coefficient ↓	Std. Coefficient
Intercept	27.376	22.679
RM	3.875	2.807
RAD	0.224	1.950
INDUS	0.093	0.649
AGE	0.021	0.594
TAX	-0.013	-2.121
CRIM	-0.080	-0.649
LSTAT	-0.612	-4.512
PTRATIO	-1.047	-2.257
NOX	-1.980	-0.230

WEKA Model Deployment

- Save model to file .model
- WEKA API



Py{WEKA}3

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier

Choose **J48 -C 0.25 -M 5**

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation Folds

☐ Percentage split %

(Nom) MF

Result list (right-click for options)

- 13:53:28 - meta.CVParameterSelection
- 14:02:53 - trees.J48

- View in main window
- View in separate window
- Save result buffer
- Delete result buffer(s)
- Load model
- Save model
- Re-evaluate model on current test set
- Re-apply this model's configuration
- Visualize classifier errors
- Visualize tree
- Visualize margin curve
- Visualize threshold curve

Classifier output

Size of the tree : 72

Time taken to build model: 0.05 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	9832	98.32	%
Incorrectly Classified Instances	168	1.68	%
Kappa statistic	0.7115		
Mean absolute error	0.0256		
Root mean squared error	0.1231		
Relative absolute error	38.9614	%	
Root relative squared error	68.0488	%	
Total Number of Instances	10000		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Ar
Weighted Avg.	0.995	0.363	0.987	0.995	0.991	0.718	0.896	0.993
	0.637	0.005	0.828	0.637	0.720	0.718	0.896	0.689
	0.983	0.351	0.982	0.983	0.982	0.718	0.896	0.982

=== Confusion Matrix ===

a	b	←-- classified as
9616	45	a = Normal
123	216	b = Failure

x 0

WEKA Model Deployment

- ตัวอย่าง
code

```
1 package weka_model;
2
3 import weka.core.*;
4 import weka.classifiers.trees.J48;
5 import weka.core.converters.ConverterUtils.DataSource;
6
7 public class Model {
8
9     public static void main(String[] args) throws Exception {
10
11         // Load Model
12         J48 j48tree = new J48();
13         j48tree = (J48) weka.core.SerializationHelper.read("<Path to .model file> PMj48.model");
14
15         // Read Unlabeled Data from source
16         DataSource source= new DataSource("<Path to .arff file> MF-test.arff");
17         Instances unlabeled=source.getDataSet() ;
18         // Set last column to Class
19         unlabeled.setClassIndex(unlabeled.numAttributes()-1);
20
21         // Classify Unlabeled data
22         System.out.println("J48 -C 0.25 -M 5");
23         for (int i = 0; i < unlabeled.numInstances(); i++) {
24             double PredictVal = j48tree.classifyInstance(unlabeled.instance(i));
25             System.out.println(unlabeled.instance(i).toString()+ " -> " + PredictVal);
26         }
27
28         // Print Option and Tree graph
29         System.out.println(j48tree.getOptions());
30         System.out.println(j48tree.graph());
31
32
33         System.out.println("\nOK Done");
34     }
35 }
36
--
```

```
MF-test.arff
1 @relation MF-test
2
3 @attribute Type {M,L,H}
4 @attribute 'Air temperature [K]' numeric
5 @attribute 'Process temperature [K]' numeric
6 @attribute 'Rotational speed [rpm]' numeric
7 @attribute 'Torque [Nm]' numeric
8 @attribute 'Tool wear [min]' numeric
9 @attribute MF {Normal,Failure}
10
11 @data
12 L,300.8,311.4,1322,67.2,18,Failure
13 L,300.7,311.9,1335,57.1,194,Failure
14 M,300.4,311.8,1553,33.1,209,Failure
15 L,302.2,310.6,1462,43.6,0,Normal
16 M,302.1,310.6,1517,38.9,2,Normal
17 L,302.1,310.6,1385,53.6,5,Normal
18
```

WEKA Model Deployment

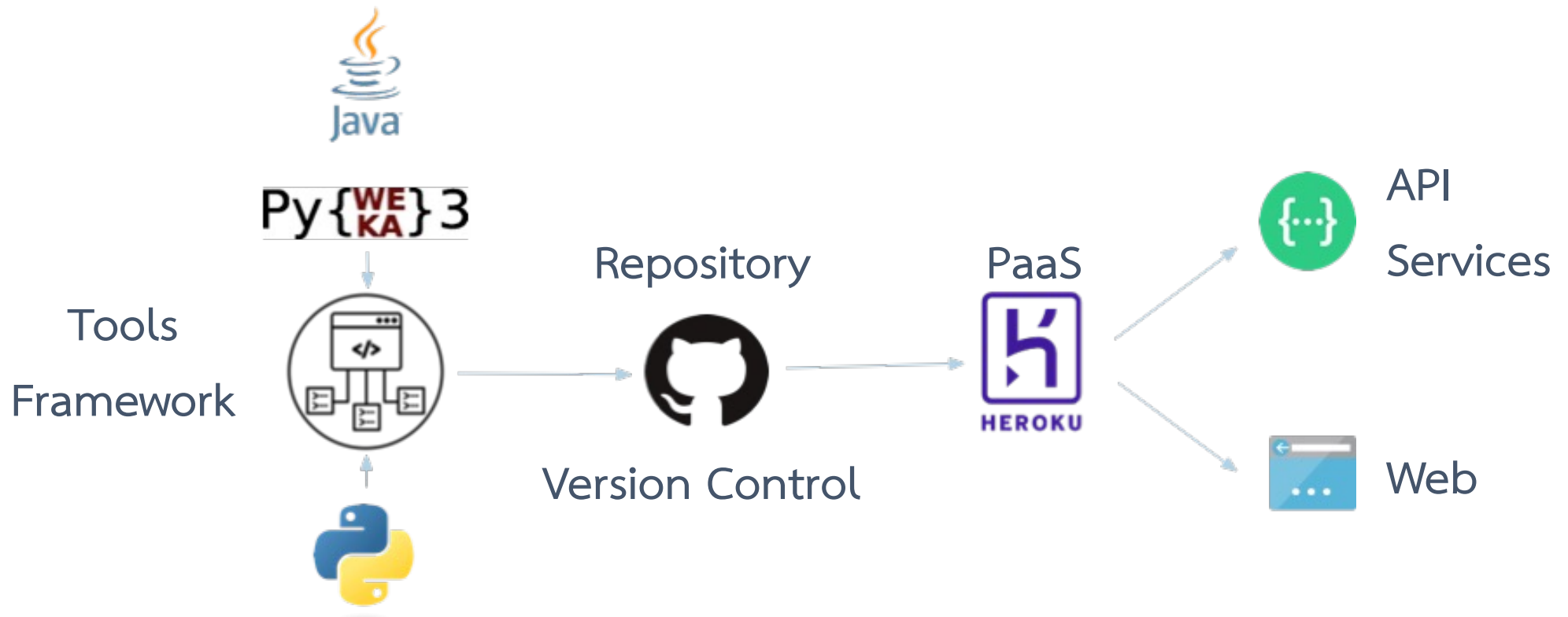
```
MF-test.arff x
1 @relation MF-test
2
3 @attribute Type {M,L,H}
4 @attribute 'Air temperature [K]' numeric
5 @attribute 'Process temperature [K]' numeric
6 @attribute 'Rotational speed [rpm]' numeric
7 @attribute 'Torque [Nm]' numeric
8 @attribute 'Tool wear [min]' numeric
9 @attribute MF {Normal,Failure}
10
11 @data
12 L,300.8,311.4,1322,67.2,18,Failure
13 L,300.7,311.9,1335,57.1,194,Failure
14 M,300.4,311.8,1553,33.1,209,Failure
15 L,302.2,310.6,1462,43.6,0,Normal
16 M,302.1,310.6,1517,38.9,2,Normal
17 L,302.1,310.6,1385,53.6,5,Normal
18
```

```
J48 -C 0.25 -M 5
L,300.8,311.4,1322,67.2,18,Failure --> 1.0
L,300.7,311.9,1335,57.1,194,Failure --> 1.0
M,300.4,311.8,1553,33.1,209,Failure --> 0.0
L,302.2,310.6,1462,43.6,0,Normal --> 0.0
M,302.1,310.6,1517,38.9,2,Normal --> 0.0
L,302.1,310.6,1385,53.6,5,Normal --> 0.0
[Ljava.lang.String;@192d3247
digraph J48Tree {
N0 [label="Torque [Nm]" ]
N0->N1 [label="<= 53.5"]
N1 [label="Torque [Nm]" ]
N1->N2 [label="<= 13"]
N2 [label="Failure (30.0/3.0)" shape=box style=filled ]
N1->N3 [label="> 13"]
N3 [label="Tool wear [min]" ]
N3->N4 [label="<= 204"]
N4 [label="Rotational speed [rpm]" ]
N4->N5 [label="<= 1379"]
N5 [label="Air temperature [K]" ]
N5->N6 [label="<= 301.5"]
N6 [label="Normal (524.0)" shape=box style=filled ]
N5->N7 [label="> 301.5"]
N7 [label="Process temperature [K]" ]
```

Topic

- Deployment Concept
- MLOps
- WEKA Model Deployment
- **Production Model Deployment**
- Rapidminer Model Deployment
- Data-Centric AI

Production Model Deployment: Local Example



Production Model Deployment: Cloud Based

colab



HUGGING FACE

 gradio



Weights & Biases



HEROKU



Streamlit



Amazon SageMaker



Google Cloud Platform



Azure Machine Learning

mlflow™

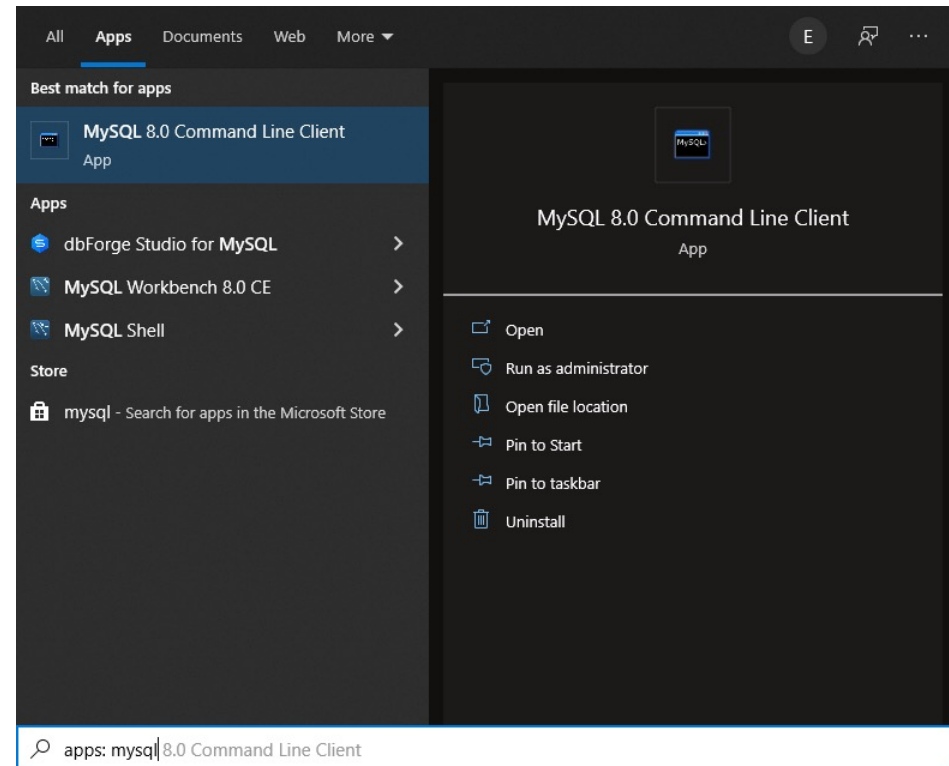
Topic

- Deployment Concept
- MLOps
- WEKA Model Deployment
- Production Model Deployment
- **Rapidminer Model Deployment**
- Data-Centric AI

Rapidminer Model Deployment

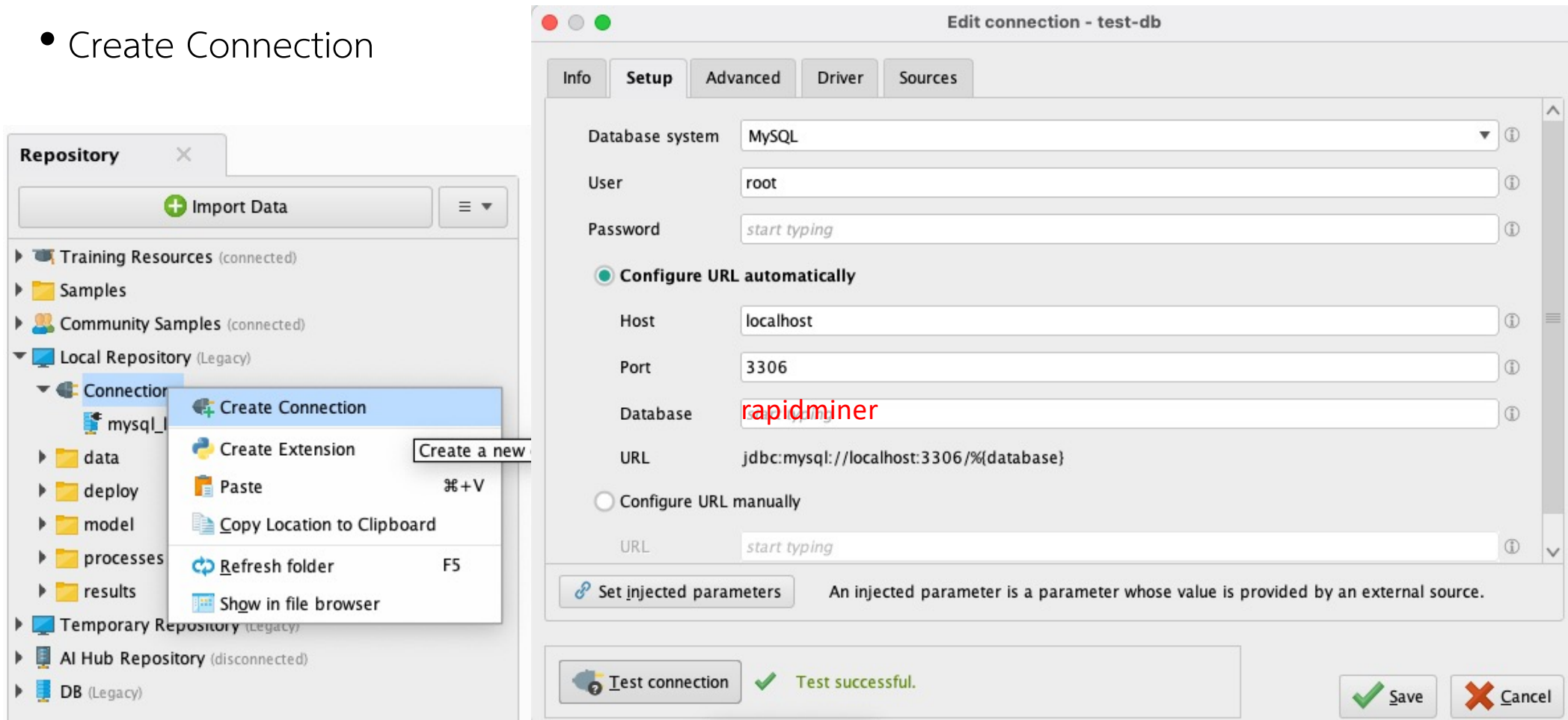
- Create Database in [MySQL 8.0 Command Line Client](#)

- Comand
CREATE DATABASE rapidminer;



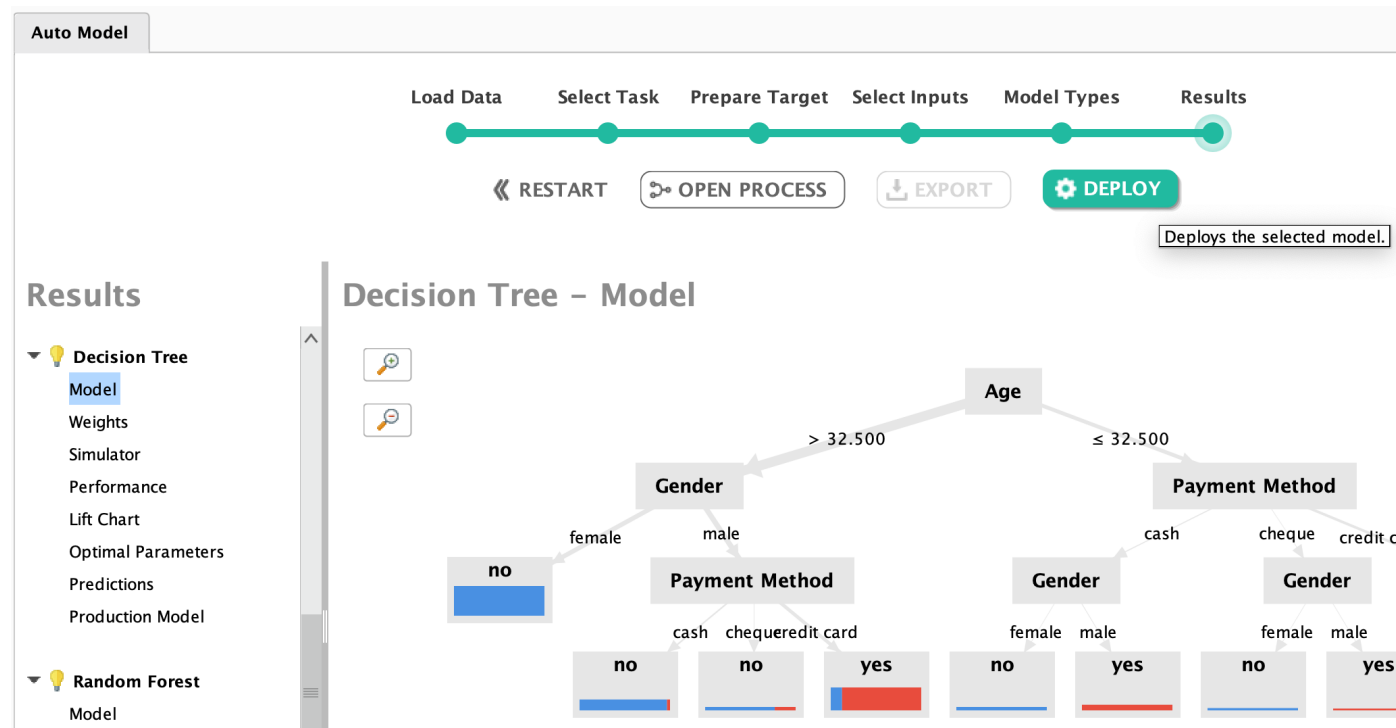
Rapidminer Model Deployment

- Create Connection



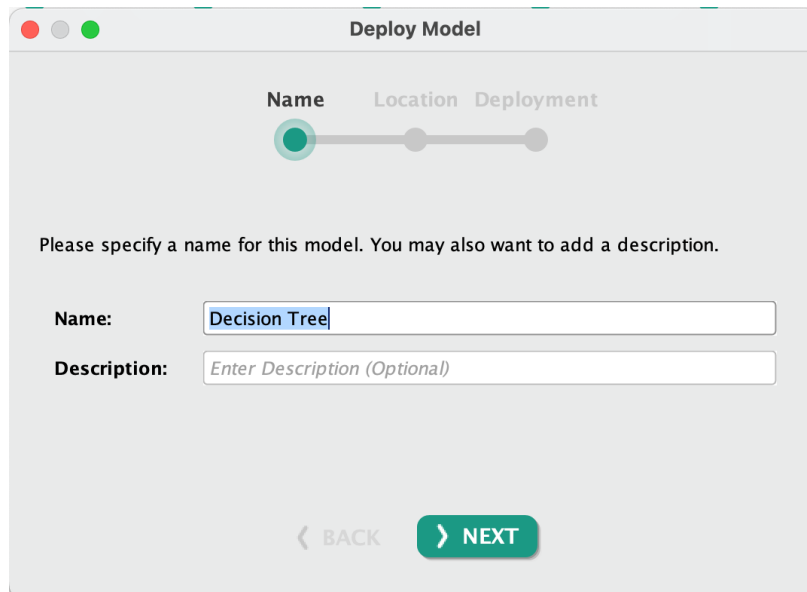
Rapidminer Model Deployment

- Deploy Deals Auto model Result



Rapidminer Model Deployment

- Deploy Deals Auto model Result



The 'Deploy Model' dialog box features a progress bar with three steps: 'Name' (active), 'Location', and 'Deployment'. Below the progress bar, a text prompt asks the user to specify a name and optionally a description. The 'Name' field contains 'Decision Tree' and the 'Description' field is empty with a placeholder 'Enter Description (Optional)'. At the bottom, there are 'BACK' and 'NEXT' buttons.

Deploy Model

Name Location Deployment

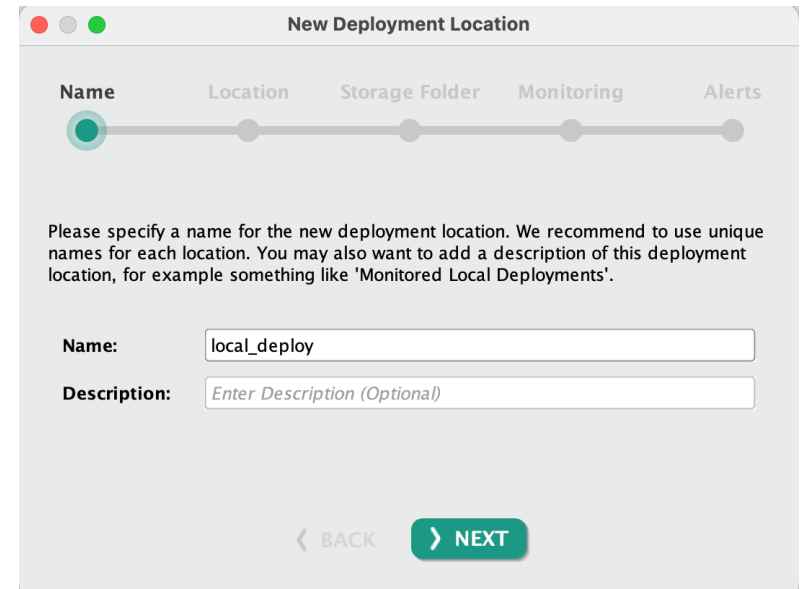
Please specify a name for this model. You may also want to add a description.

Name:

Description:

< BACK **> NEXT**

Add New Deployment



The 'New Deployment Location' dialog box features a progress bar with five steps: 'Name' (active), 'Location', 'Storage Folder', 'Monitoring', and 'Alerts'. Below the progress bar, a text prompt asks the user to specify a name and optionally a description. The 'Name' field contains 'local_deploy' and the 'Description' field is empty with a placeholder 'Enter Description (Optional)'. At the bottom, there are 'BACK' and 'NEXT' buttons.

New Deployment Location

Name Location Storage Folder Monitoring Alerts

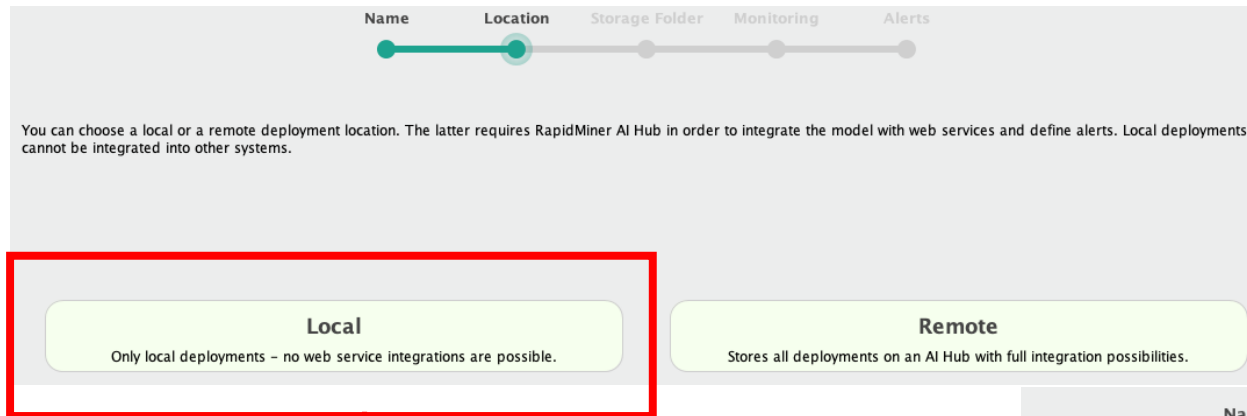
Please specify a name for the new deployment location. We recommend to use unique names for each location. You may also want to add a description of this deployment location, for example something like 'Monitored Local Deployments'.

Name:

Description:

< BACK **> NEXT**

Rapidminer Model Deployment



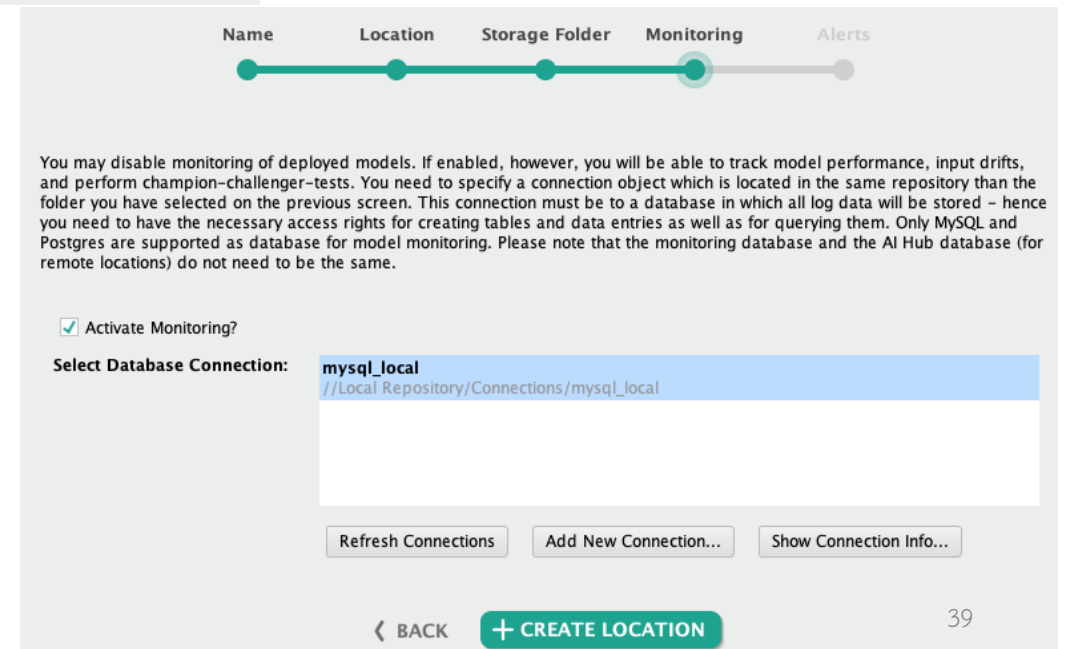
This screen shows the 'Location' step in a five-step deployment process. The steps are Name, Location, Storage Folder, Monitoring, and Alerts. The 'Location' step is currently active. Below the progress bar, a text box explains that users can choose between local and remote deployment. Local deployment is highlighted with a red box, indicating it is the selected option.

Name **Location** Storage Folder Monitoring Alerts

You can choose a local or a remote deployment location. The latter requires RapidMiner AI Hub in order to integrate the model with web services and define alerts. Local deployments cannot be integrated into other systems.

Local
Only local deployments – no web service integrations are possible.

Remote
Stores all deployments on an AI Hub with full integration possibilities.



This screen shows the 'Monitoring' step in the deployment process. The 'Monitoring' step is active. It provides instructions on how to enable or disable monitoring and how to select a database connection. A checkbox for 'Activate Monitoring?' is checked. Below it, a list of database connections is shown, with 'mysql_local' selected. At the bottom, there are buttons for 'Refresh Connections', 'Add New Connection...', and 'Show Connection Info...'. A red arrow from the 'Local' option in the previous screen points to this screen.

Name **Location** Storage Folder **Monitoring** Alerts

You may disable monitoring of deployed models. If enabled, however, you will be able to track model performance, input drifts, and perform champion–challenger–tests. You need to specify a connection object which is located in the same repository than the folder you have selected on the previous screen. This connection must be to a database in which all log data will be stored – hence you need to have the necessary access rights for creating tables and data entries as well as for querying them. Only MySQL and Postgres are supported as database for model monitoring. Please note that the monitoring database and the AI Hub database (for remote locations) do not need to be the same.

☒ Activate Monitoring?

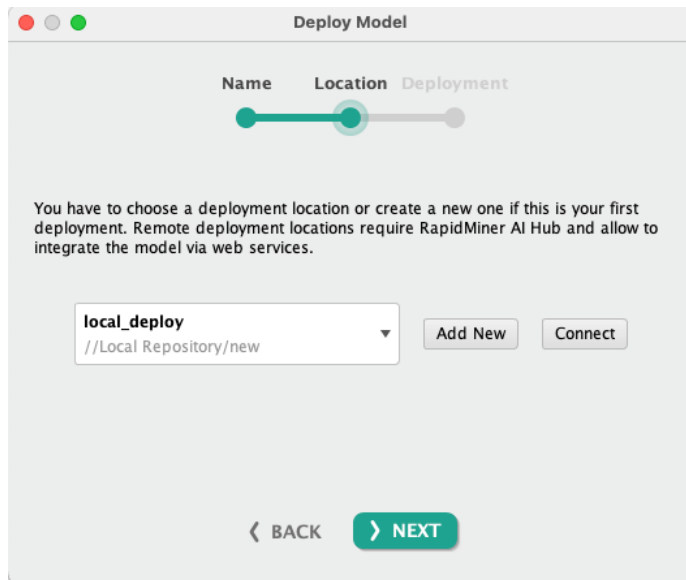
Select Database Connection:

mysql_local
//Local Repository/Connections/mysql_local

Refresh Connections Add New Connection... Show Connection Info...

◀ BACK + CREATE LOCATION

Rapidminer Model Deployment



Deploy Model

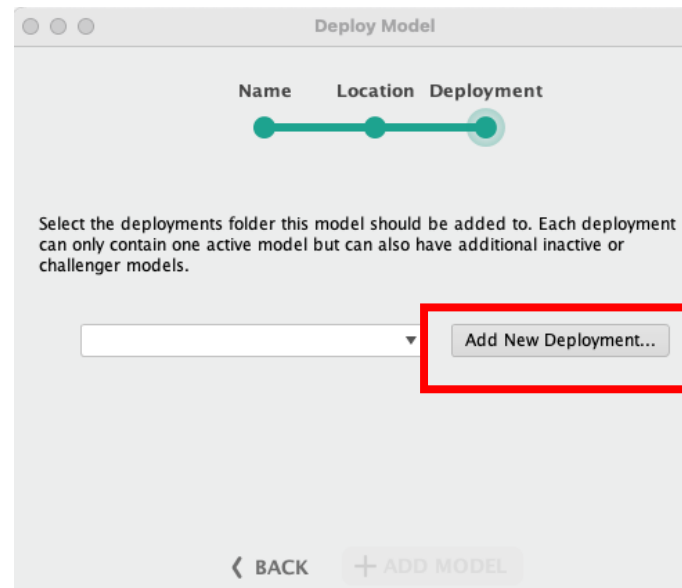
Name Location Deployment

You have to choose a deployment location or create a new one if this is your first deployment. Remote deployment locations require RapidMiner AI Hub and allow to integrate the model via web services.

local_deploy
//Local Repository/new

Add New Connect

< BACK > NEXT



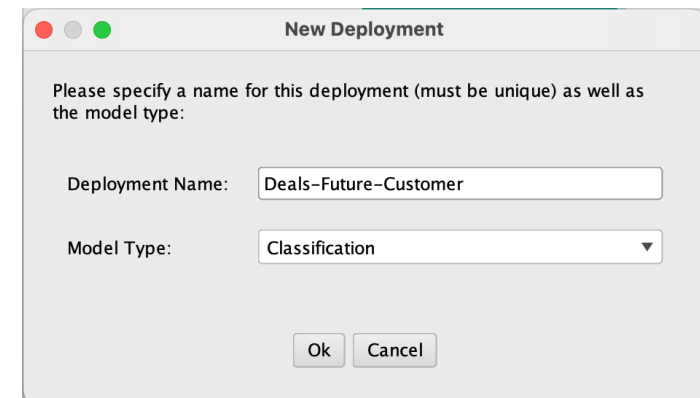
Deploy Model

Name Location Deployment

Select the deployments folder this model should be added to. Each deployment can only contain one active model but can also have additional inactive or challenger models.

Add New Deployment...

< BACK + ADD MODEL



New Deployment

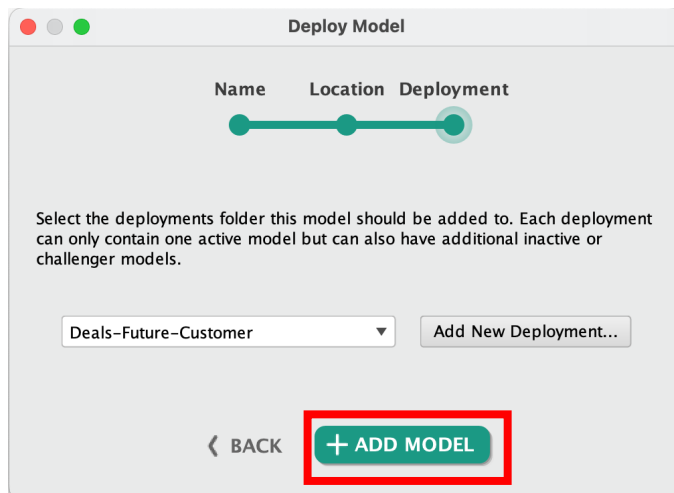
Please specify a name for this deployment (must be unique) as well as the model type:

Deployment Name: Deals-Future-Customer

Model Type: Classification

Ok Cancel

Rapidminer Model Deployment



- Then add other models

Deployments

Deals-Future-Customer: Models

Shows all models in this deployment and allows to activate models or use them as challengers. ⓘ

DASHBOARD MODELS PERFORMANCE DRIFTS SIMULATOR SCORING ALERTS INTEGRATIONS

DEPLOY AUTO MODEL DEPLOY CUSTOM MODEL

Name	Type	Created	Author	Status	Predeployment Error	Recent Error	Scoring Time (1,000 Rows)
Deep Learning	Deep Learning	Dec 4, 2022 1:16...	Unknown	Active	2.1% ± 1.5%	4.3%	104 ms
Decision Tree	Decision Tree	Dec 4, 2022 1:17...	Unknown	Challenger	7.0% ± 2.7%	8.1%	44 ms
Logistic Regression	Logistic Regression	Dec 4, 2022 1:16...	Unknown	Challenger	2.8% ± 1.5%	4.1%	98 ms
Random Forest	Random Forest	Dec 4, 2022 1:17...	Unknown	Challenger	3.1% ± 1.4%	4.1%	2 s

After Scoring

Rapidminer Model Scoring

Deployments

Deals-Future-Customer: Scoring

Upload data to score it or upload actual outcomes for performance calculations.

DASHBOARD

MODELS

PERFORMANCE

DRIFTS

SIMULATOR

SCORING

ALERTS

INTEGRATIONS

SCORE DATA

DEFINE ACTUALS

Apply your model

You do not have created any prediction yet.

Want to learn more? Watch a quick video.

EXPORT

Score Data

Select Data

Define Columns

Apply Model

Please select a data set on which the active model in this deployment will be applied. It should be of the same structure as the original training data you have provided, i.e. should have the same columns and column types. If columns are missing, they will be created with all values missing and will be ignored. You can, however, define a column containing the true labels (if available) in the next step.

Training Resources (connected)

Samples

data

Deals

Deals-Testset

Golf

Golf-Testset

BACK

NEXT

Score Data

Select Data

Define Columns

Apply Model

You see the data you have selected below. Columns which have been an input factor in the training data are shown in orange. If one of the columns should be treated as one containing the actual outcomes or as an ID column, you can specify those columns by selecting those roles in the context menu shown at a right-click on those columns. In certain cases, RapidMiner will suggest those columns for you as well. Please make sure that those suggestions are correct then.

Target Column

Age Number	Gender Category	Payment Method Category	Future Customer Category
44	male	credit card	yes
86	female	credit card	no
17	female	credit card	yes
57	female	credit card	no
48	female	credit card	no

BACK

NEXT

Score Data

Select Data

Define Columns

Apply Model

Below is a summary of your choices. If everything is as desired, press the Score Data button and RapidMiner will calculate the predictions for the provided data using the active model in this deployment. This may take a while.

Special Columns

If you have identified a column containing the actual outcomes for the original target column or which should be used to identify the cases, they will be shown below:

Actual Outcomes for Target: Future Customer

Used Columns

These columns are matching columns from the training data and will be used as they are:

Age

Gender

Payment Method

BACK

SCORE DATA

42

Rapidminer Model Scoring

Deployments

Deals-Future-Customer: Scoring

Upload data to score it or upload actual outcomes for performance calculations. ⓘ

DASHBOARD MODELS PERFORMANCE DRIFTS SIMULATOR **SCORING** ALERTS INTEGRATIONS

SCORE DATA

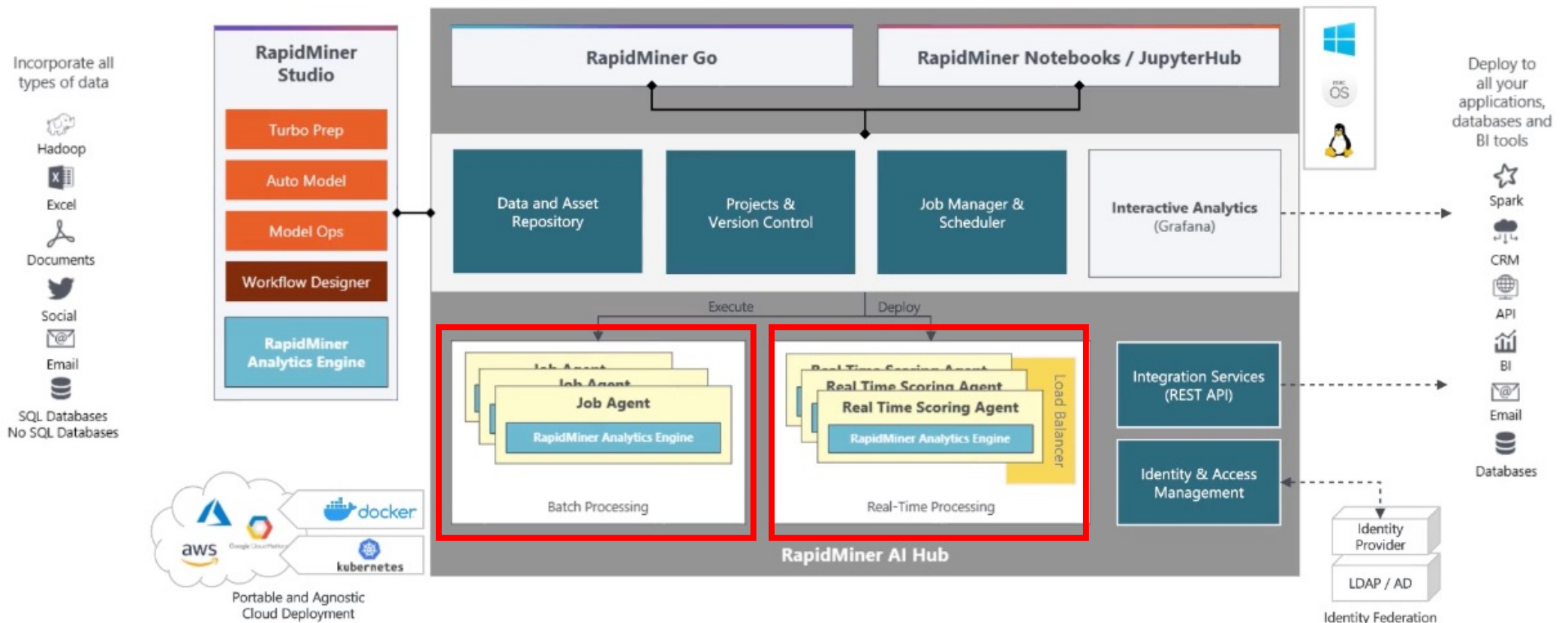
DEFINE ACTUALS

Error Rate: 1.00%

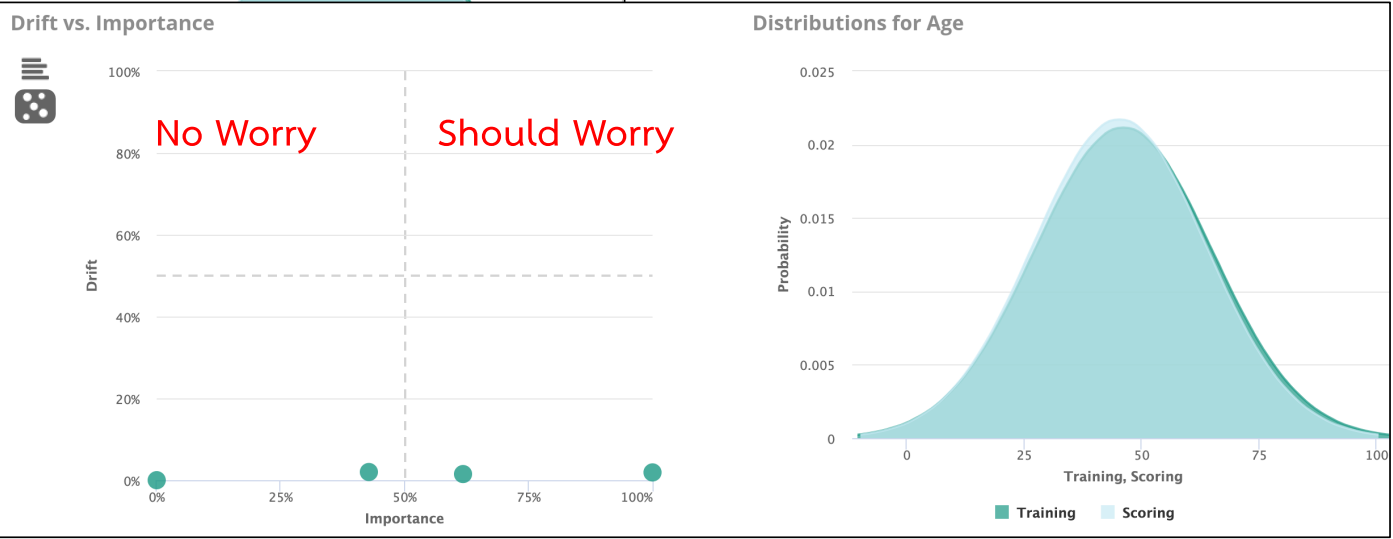
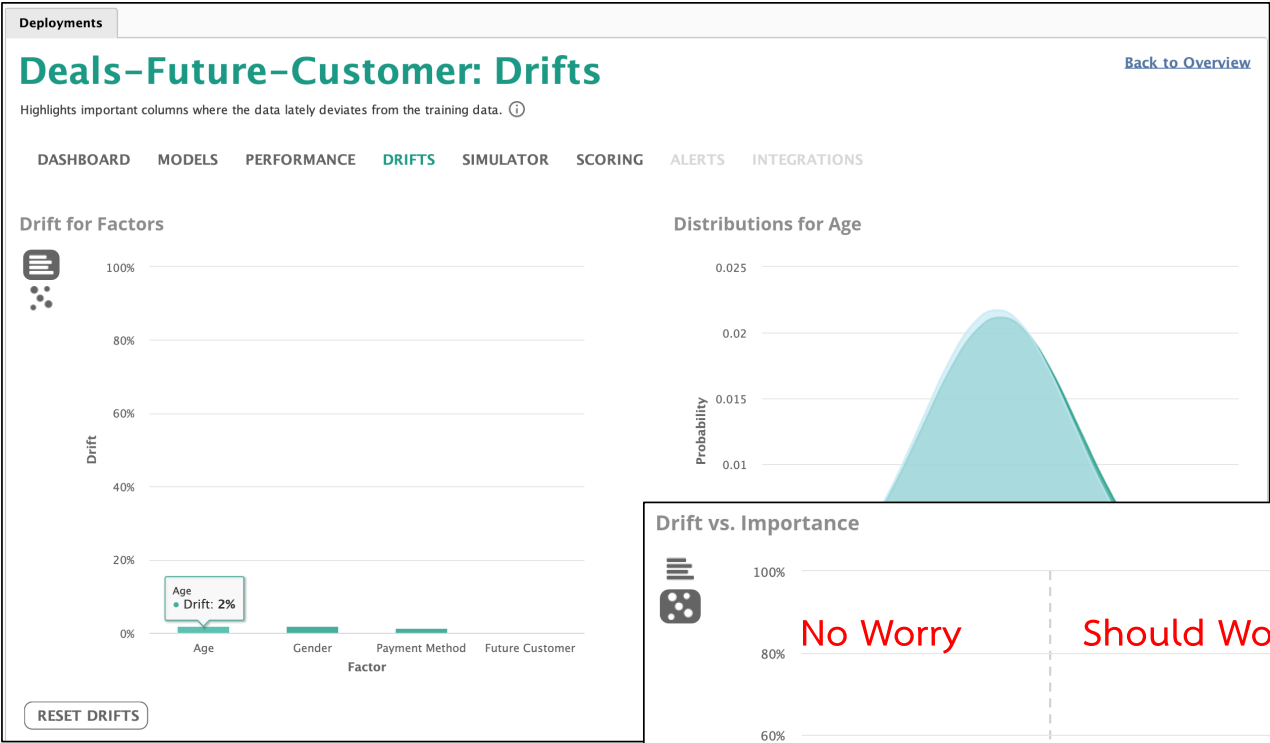
Row No.	Future Cus...	prediction(...	confidence(no)	confidence(yes)	cost	Gender	Payment M...	Age
1	yes	yes	0	1	1	male	credit card	44
2	no	no	0.833	0.167	0.667	female	credit card	86
3	yes	yes	0	1	1	female	credit card	17
4	no	no	0.833	0.167	0.667	female	credit card	57
5	no	no	0.833	0.167	0.667	female	credit card	48
6	no	no	0.833	0.167	0.667	female	credit card	39
7	no	no	0.833	0.167	0.667	male	credit card	78
8	yes	yes	0	1	1	male	credit card	38
9	no	no	0.833	0.167	0.667	male	cheque	80
10	yes	yes	0	1	1	male	credit card	33
11	yes	yes	0	1	1	male	credit card	54

EXPORT

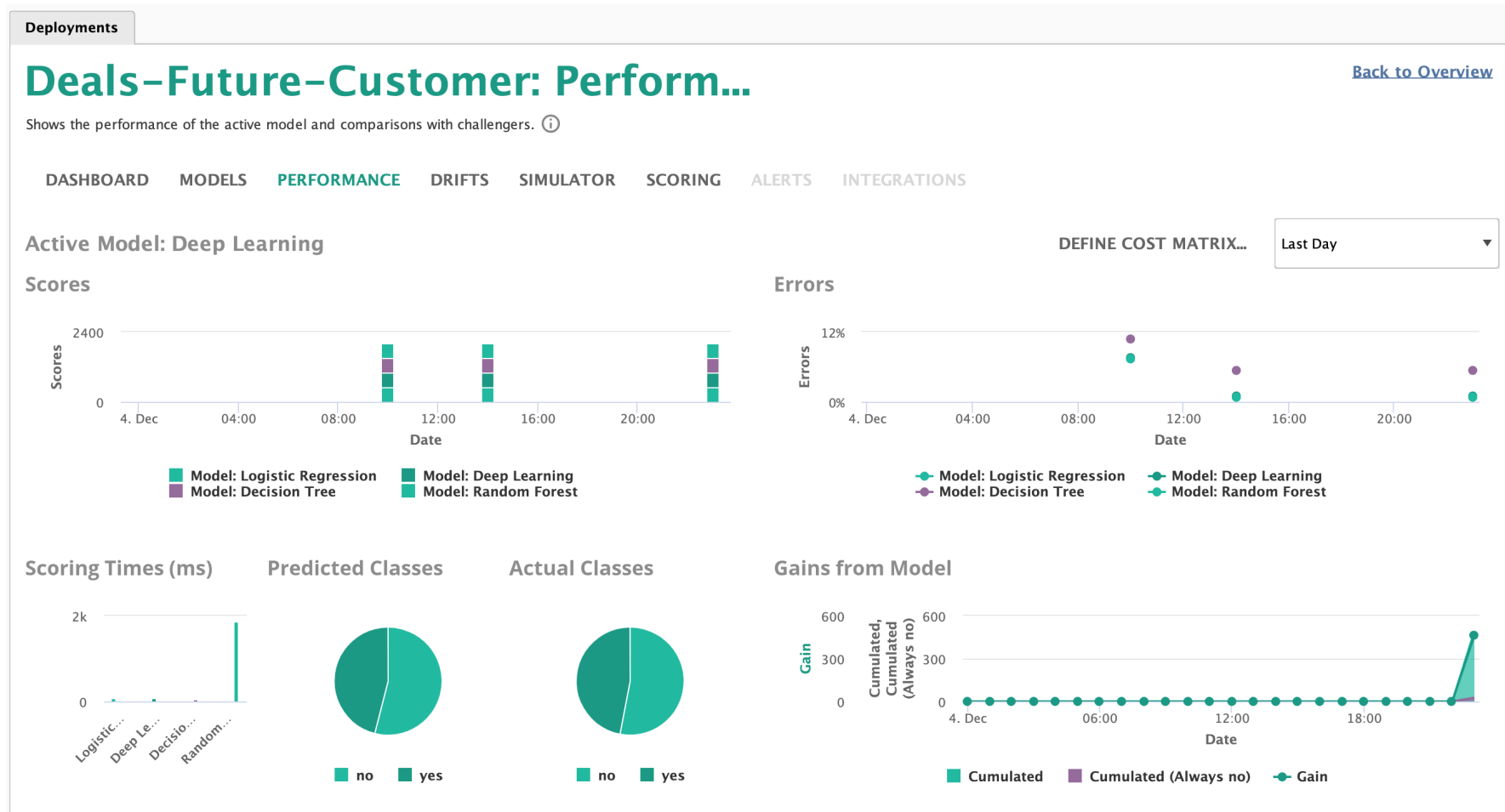
Rapidminer Platform



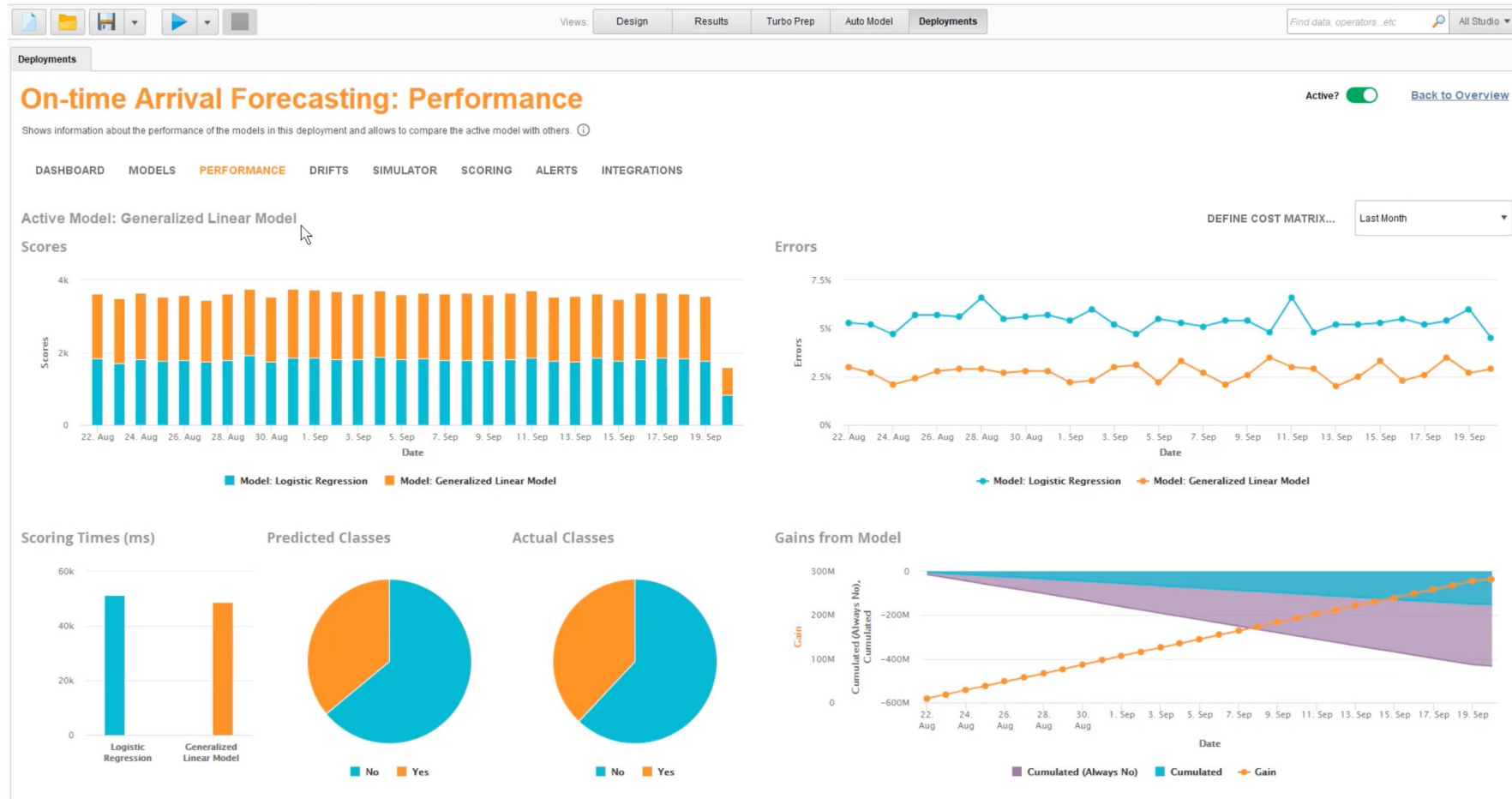
Drifts



Rapidminer Model Performance Monitoring



Example Model Performance Monitoring



<https://academy.rapidminer.com/learn/video/rapidminer-model-operations-management>

Topic

- Deployment Concept
- MLOps
- WEKA Model Deployment
- Production Model Deployment
- Rapidminer Model Deployment
- **Data-Centric AI**

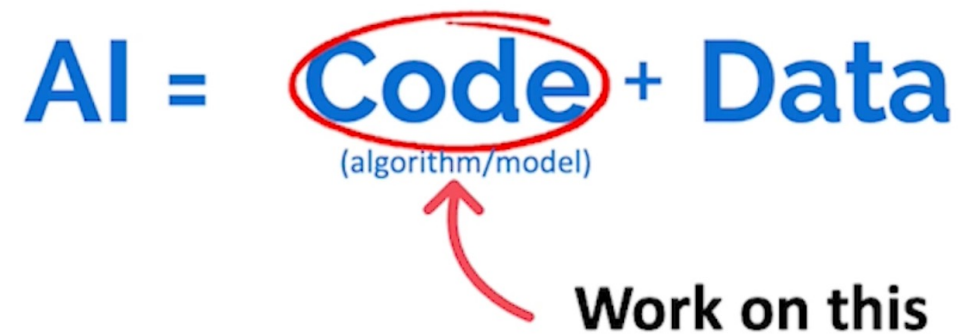
Model-Centric AI

Model-centric approach

$$\text{AI} = \text{Code} + \text{Data}$$

(algorithm/model)

Work on this

The diagram shows the equation 'AI = Code + Data' in blue text. The word 'Code' is circled in red. Below 'Code' is the text '(algorithm/model)' in a smaller blue font. A red curved arrow points from the text 'Work on this' to the word 'Code'.

Data-Centric AI

Data-centric approach

$$\text{AI} = \text{Code} + \text{Data}$$

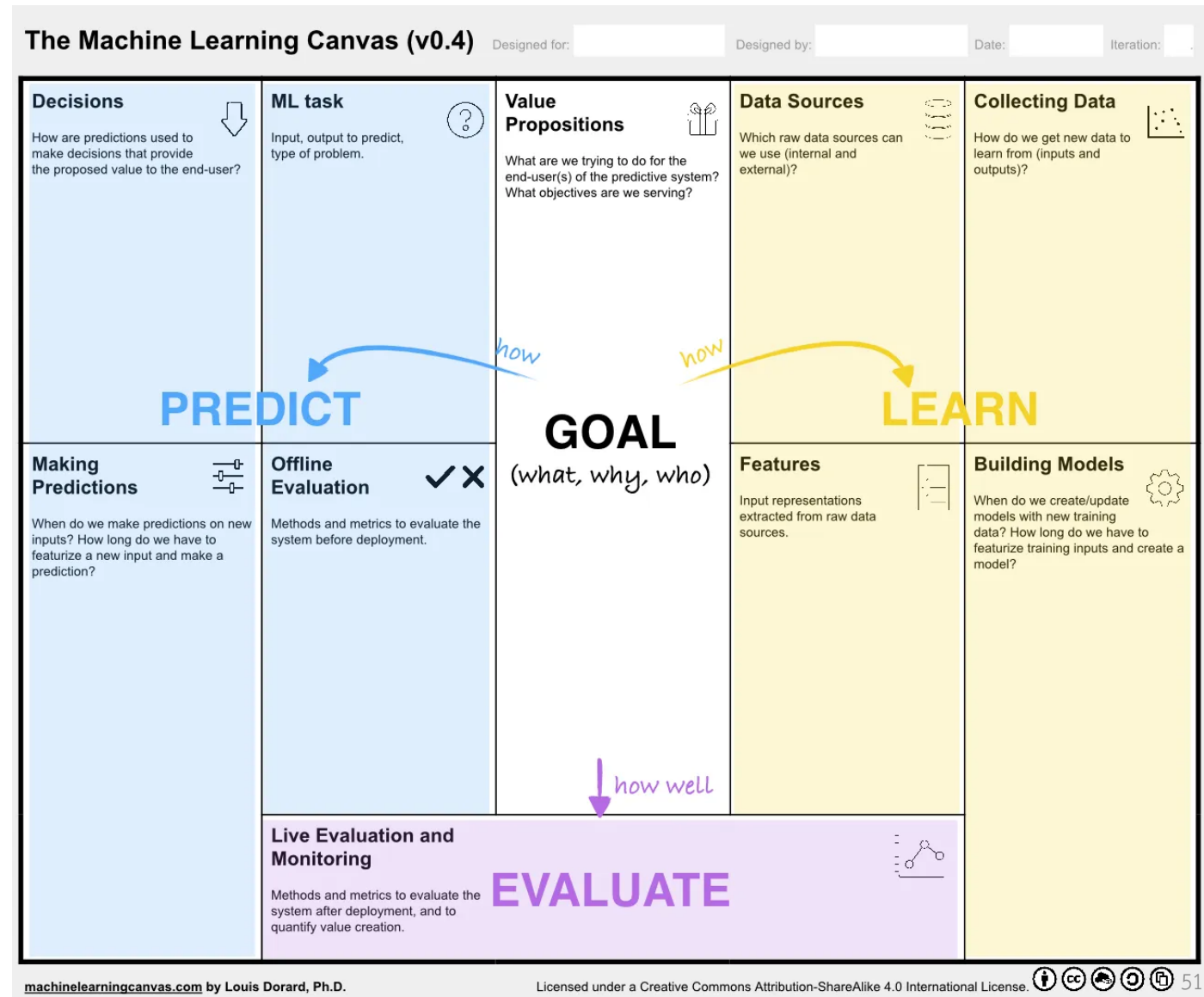
(algorithm/model)

The diagram shows the equation 'AI = Code + Data' in blue text. The word 'Code' has '(algorithm/model)' written below it in a smaller blue font. The word 'Data' is circled in red. A red arrow points from the text 'Work on this' below to the circled 'Data'.

Work on this

AI/ML Project Revisited

- <https://www.machinelearningcanvas.com/>



Q & A

Ref:

